Computer and robot vision

Course notes

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1 Image formation

A camera is the union of the sensors and the software that uses the image to determine relevant points in the image plane and the projection model. This allows to calculate the lines connecting the points in the image plane and the object.

> A camera sensor (once calibrated) does not measure points in the image, but measures lines in the space (a subset of the star of lines passing trough the pinhole).

Since we are only calculating lines, knowing the precise physical measures of the image plane is not necessary since the lines don't depend on them. The simplest geometric model of image formation is the **pinhole camera**, it is based on the same principle as the camera obscura.

1.1 Pinhole camera model

The tiny hole (pinhole) in one wall of the room lets in a ray of light for each point in the scene so that an image of the outside world is drawn on the opposite wall (**image plane**).

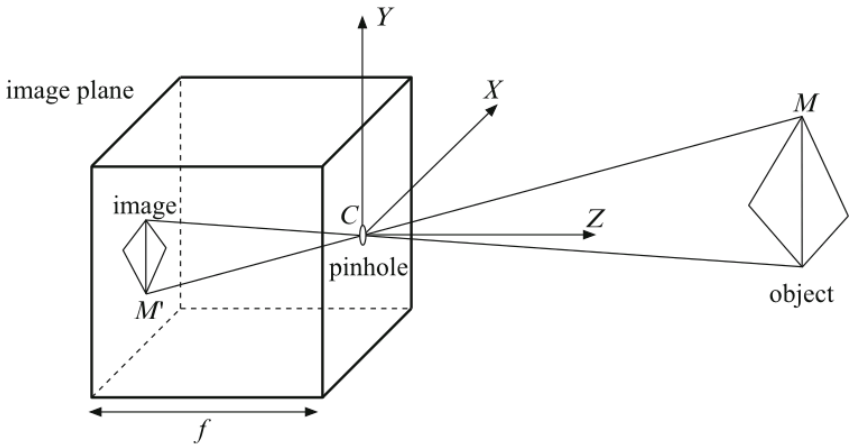


Figure 1: Image formation in the camera obscura/pinhole camera

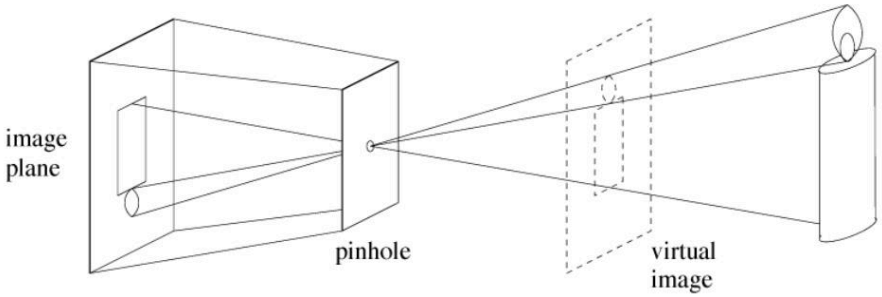


Figure 2: Pinhole camera with virtual image

The geometric model of the camera that we will adopt is the so-called pinhole or perspective model, which consists (refer to Figure 3) of the **image plane** Q and the **centre of Projection (COP)** C, located at a distance f (**focal length**) from the image plane.

> **Definition 1.1** *(principal axis).* The line through C orthogonal to Q is the **principal axis** of the camera

> **Definition 1.2** *(principal point).* The intersection of the principal axis with Q is called the **principal point** (or center of the image).

Definition 1.3 *(focal plane).* The plane F parallel to Q and containing C is called the **focal plane**.

The points in F are projected to infinity on Q.

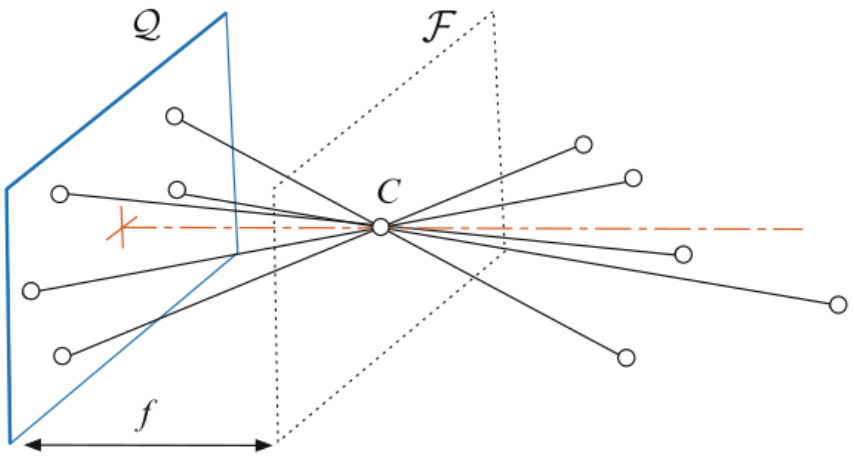


Figure 3: Pinhole camera with virtual image

Let M be a point in the scene, of coordinates (X, Y, Z), and let M' of coordinates (X', Y', Z') be its projection onto the image plane through the pinhole C. If f is the distance of C from the image plane (focal length), then from the similitude of the triangles we get:

$$\frac{-X'}{f} = \frac{X}{Z} \qquad \frac{-Y'}{f} = \frac{Y}{Z} \tag{1}$$

therefore

$$X' = \frac{-fX}{Z} \qquad Y' = \frac{-fY}{Z} \qquad Z' = -f \tag{2}$$

Note that the image is inverted with respect to the scene, both left-right and top-bottom, as indicated by the minus sign. These equations define the image formation process, which is called **perspective projection**.

The parameter f determines the **magnification factor** of the image: if the image plane is close to the pinhole, the image is reduced in size, while it will become larger and larger as f increases. Since the working area of the image plane is framed, moving the image plane away also reduces the **field of view**, that is the solid angle that contains the portion of the scene that appears in the image.

The division by Z is responsible for the foreshortening effect, whereby the apparent size of an object in the image decreases according to its distance from the observer.

1.1.1 Simplified pinhole model

The coordinate system in which the points of the three-dimensional space are represented is called the **world (or object) coordinate system**. Let us introduce the **camera reference frame**, a right-handed triplet (X, Y, Z) of axes centred in C and with the Z axis coincident with the principal axis. In this simplified model, the world reference frame is chosen coincident with the camera reference frame.

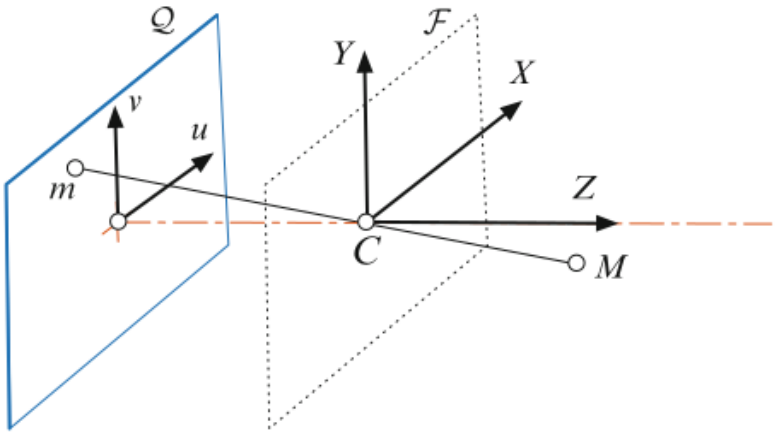


Figure 4: Simplified model of the camera with reference frames. The point M is projected onto the point m where the line CM intersects the image plane Q

We also introduce a **reference frame** (u, v) for the plane Q centred at the principal point and with the axes u and v oriented as X and Y , respectively, as shown in Figure 4.

Unless stated otherwise, we use ~ to denote Cartesian coordinates to distinguish them from homogeneous ones.

Now consider a point M of coordinates $\widetilde{M} = \begin{pmatrix} X \\ Y \\ Z \end{pmatrix}$ in 3D space and let m of coordinates $\tilde{m} = \begin{pmatrix} u \\ v \end{pmatrix}$ be its projection onto Q trough C. We have that

$$\begin{cases} u = \frac{-f}{Z} X \\ v = \frac{-f}{Z} Y \end{cases} \tag{3}$$

In Equation 3 we have the intersection of two planes, therefore we obtain a line. This is the **perspective projection**. The transformation in Cartesian coordinates is clearly non-linear because of the division by Z. But if we use homogeneous coordinates, the transformation becomes linear. Let

$$m = \begin{pmatrix} u \\ v \\ 1 \end{pmatrix} \quad M = \begin{pmatrix} X \\ Y \\ Z \\ 1 \end{pmatrix} \tag{4}$$

be the homogeneous coordinates of m and M, respectively. So the equation for perspective projection, in this simplified case, rewrites:

$$Z \begin{pmatrix} u \\ v \\ 1 \end{pmatrix} = Z \begin{pmatrix} -f\frac{X}{Z} \\ -f\frac{Y}{Z} \\ Z \end{pmatrix} = \underbrace{\begin{pmatrix} -f & 0 & 0 & 0 \\ 0 & -f & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}}_{P} \begin{pmatrix} X \\ Y \\ Z \\ 1 \end{pmatrix} \tag{5}$$

In other words

$$m = \frac{1}{Z} PM \tag{6}$$

The matrix P represents the geometric model of the camera and is called **Perspective Projection Matrix (PPM)** or **camera matrix**.

1.1.2 General pinhole model

A realistic model of a camera, describing the transformation from 3D coordinates to pixel coordinates, as well as the perspective transformation, must take into account:

- The change of coordinates from metres in the sensor plane to pixels in the image.
- The change of coordinates between the camera reference frame and the world reference frame.

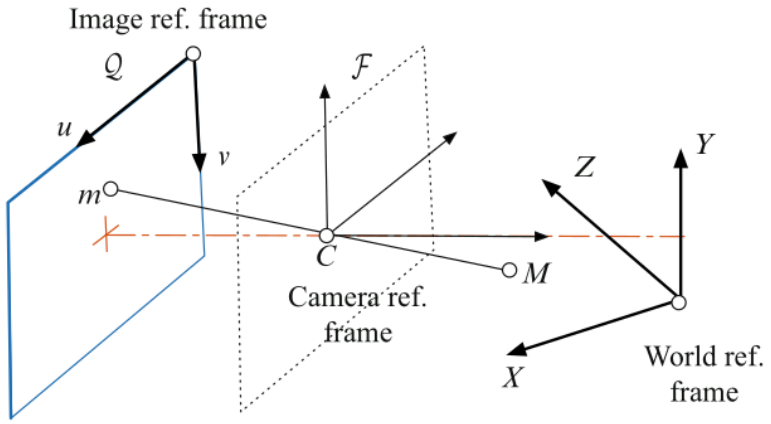
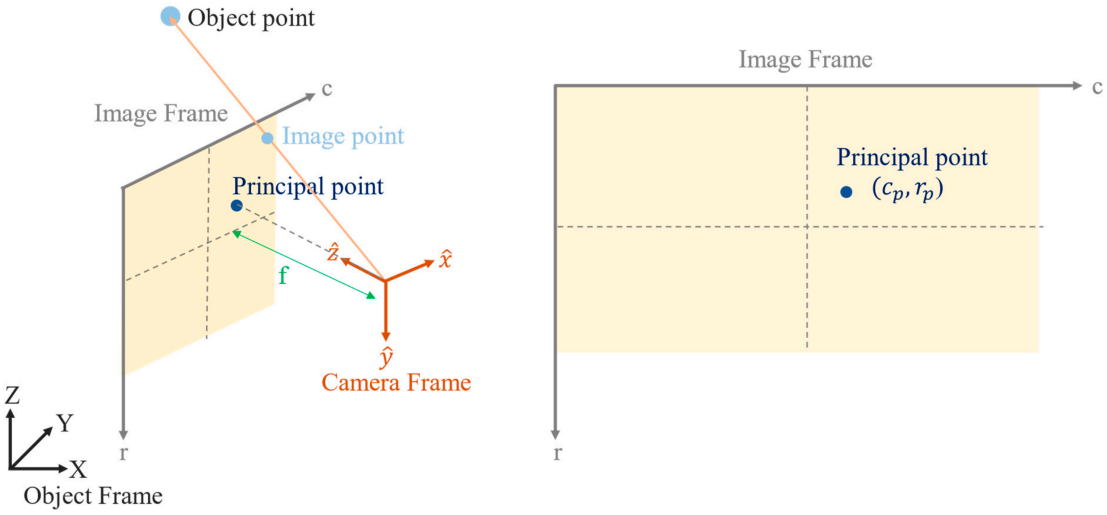


Figure 5: General model of the camera with reference frames. Note that the image plane should be viewed from the half-space opposite to C, so that the origin of the image reference is in the upper left corner of the image



(a) (b)

Figure 6: Geometry of perspective projection: (a) the three coordinate frames involved and (b) the image coordinates of the principal point.

1.1.3 Intrinsic parameters

The transformation from the sensor coordinates to pixels is modelled by an affine transformation A that takes into account the **translation of the origin at the upper left corner**, the independent rescaling of the u and v axes for changing the unit of measurement from metres to pixels and the reversal of the direction of the axes:

$$A = \begin{pmatrix} -k_u & 0 & u_0 \\ 0 & -k_v & v_0 \\ 0 & 0 & 1 \end{pmatrix} \tag{7}$$

where (u_0, v_0) are the coordinates in pixels of the principal point (which is not on the vertical axis!) and $k_u(k_v)$ (which is translated in respect to the center of the lens) is the inverse of the effective pixel size along the direction of the u (v) axis, measured in m^{-1}.

The matrix A transforms the coordinates from metres to pixels; hence, it pre-multiplies the PPM (refer to Section A.1), which becomes:

$$P = \underbrace{\begin{pmatrix} -k_u & 0 & u_0 \\ 0 & -k_v & v_0 \\ 0 & 0 & 1 \end{pmatrix}}_{A} \underbrace{\begin{pmatrix} -f & 0 & 0 & 0 \\ 0 & f & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}}_{\text{PPM}} = \begin{pmatrix} fk_u & 0 & u_0 & 0 \\ 0 & fk_v & v_0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix} = K(I_3 \;\; 0) \tag{8}$$

where

$$K = \begin{pmatrix} fk_u & 0 & u_0 \\ 0 & fk_v & v_0 \\ 0 & 0 & 1 \end{pmatrix} \tag{9}$$

is called the **intrinsic parameters matrix** of the camera, while $(I_3 \;\; 0)$ is the perspective transformation without parameters. It is obtained when the world reference frame coincides with the camera reference frame and the image points are expressed in normalized image coordinates, that is,

$$q = K^{-1}m \tag{10}$$

> Note that whereas pixel coordinates are always measurable in the image, to step to normalized image coordinates the matrix of intrinsic parameters is needed.

> The multiplication by the PPM matrix calculates the projection of the point in the image plane, while the multiplication by matrix A does the translation with respect to the principal point. The multiplication by the PPM matrix returns a 3×1 matrix with the coordinates of the point in the image plane expressed in metres, while the multiplication by matrix A returns a 3×1 matrix with the coordinates of the point in pixels.

A more general K includes also the **skew** angle Θ that corresponds to the angle between the u and v axes, that normally is $\frac{\pi}{2}$:

$$K = \begin{pmatrix} fk_u & \frac{-fk_u}{\tan(\Theta)} & u_0 \\ 0 & \frac{fk_v}{\sin(\Theta)} & v_0 \\ 0 & 0 & 1 \end{pmatrix} \tag{11}$$

This matrix has five degrees of freedom, which are the intrinsic parameters of the camera: fk_u, fk_v, u_0, v_0, and Θ.

> The skew angle Θ is usually very close to $\frac{\pi}{2}$, so it is often neglected in practice.

1.1.4 Extrinsic parameters

To account for the fact that—in general—the world reference frame does not coincide with the camera reference frame, we must introduce a coordinate change consisting of a direct **isometry**, that is, a **rotation** $\boldsymbol{R}$ and a **translation** $\boldsymbol{t}$. We denote by M_c the homogeneous coordinates of a point in the camera coordinate system and M the homogeneous coordinates of the same point in the world coordinate system. We can therefore write:

$$M_c = GM \tag{12}$$

where

$$G = \begin{pmatrix} R & t \\ 0 & 1 \end{pmatrix} \tag{13}$$

The matrix G encodes the **exterior orientation** (position and angular attitude[1]) of the camera with respect to the world reference frame. Specifically, the rows of R are the X, Y and Z axes of the camera reference frame expressed in the world coordinate system.

By appropriately parameterising the rotation R, the exterior orientation of the camera is encoded by six **extrinsic parameters**:

- three Euler angles ω, φ, κ that specify the rotation
- three components t_1, t_2, t_3 for the translation.

From Equation 8 and Equation 11 we can write the general perspective projection matrix as:

$$P = K[I_3, 0]G \tag{14}$$

This is the most general form of the perspective projection matrix:

- G encodes the exterior orientation of the camera
- K contains its intrinsic parameters (by analogy, we say that it encodes the interior orientation)
- $[I_3, 0]$ represents a perspective transformation from camera coordinates to normalised image coordinates.

In total, the perspective projection matrix has **11 degrees of freedom**:

- 6 for the exterior orientation
- 5 for the interior orientation.

1.1.5 Summarization of point translation

Here we summarize the coordinate systems introduced and their relationship.

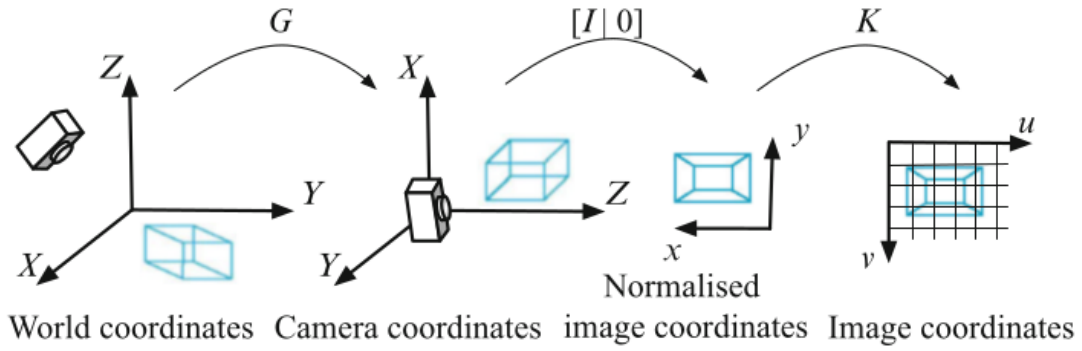


Figure 7: Coordinate systems involved in camera modelling

- **World coordinates**: generic coordinates that represent points in 3D, $\widetilde{M} = \begin{pmatrix} X \\ Y \\ Z \end{pmatrix}$
- **Camera coordinates**: coordinates of the 3D points in a reference frame with origin in the COP and integral with the camera.

They will often be referred to with the same notation as the world coordinates $\widetilde{M} = \begin{pmatrix} X \\ Y \\ Z \end{pmatrix}$, when needed to be distinguished we use $\widetilde{M}_c$.

- **Normalised image coordinates**: are measured in metres with origin at the principal point and denoted by $\tilde{q} = \begin{pmatrix} x \\ y \end{pmatrix}$.
- **Image coordinates**: are measured in pixels with origin in the upper left corner of the image and denoted by $m = \begin{pmatrix} u \\ v \end{pmatrix}$.

[1] In navigation orientation is the determination of the east cardinal point (oriens in latin); by extension, in topography and photogrammetry, it indicates the determination of position and direction with respect to cardinal points or other reference frames.

1.1.6 Thin lenses

A pinhole camera is really small (in theory, infinitesimal) therefore the energy reaching the image plane also is. In order to obtain an image we would need a transformation rate which is, in theory, infinite and therefore impossible to obtain. If we simply increased the size in the hole we would not be able to control the amount of energy coming to the image plane, and would therefore obtain a spot instead of a point the image plane. If this were to occur, then the image would be blurry (**circle of confusion**).

Vertebrate eyes and cameras use lenses. A lens, being larger than a pinhole, can gather more light. Lenses allow a larger opening, so a finite amount of energy, to reach the image plane and at the same time we might be able to obtain an effect in terms of projection (zoom, etc...). The downside is that not the whole scene can be in focus at the same time.

The approximation we make for the optics of the imaging system— which in general is very complex, being made up of multiple lenses is that of the thin lens.

Definition 1.4 *(Thin lenses).* Thin lenses are ideal devices that enjoy the following properties:

- The rays parallel to the optical axis incident on the lens are refracted so that they pass through a point on the optical axis called focus F
- The rays passing through the centre C of the lens are unaffected.

The distance of the focus F from the centre of the lens C is called **focal length** D, which depends on the radii of curvature of the two lens surfaces and the refractive index of the constituent material.

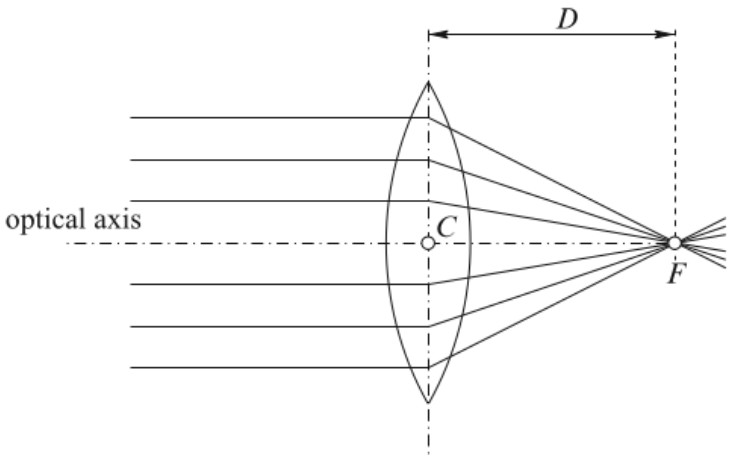


Figure 8: Thin lenses

For non-thin lenses there is no “center” of the lens, but two nodal points (front and rear) are defined to characterize its behavior. By definition, an input ray directed at the front nodal point leads to an output ray which appears to come from the back nodal point, forming the same angle with respect to the optical axis. For a thin lens, the two nodal points coincide in the centre of the lens.

Given a point of the scene M it is possible to construct graphically the image (or conjugate point) M', using two special rays that depart from M: the ray parallel to the optical axis, which after refraction, passes through F and the ray that passes unchanged through C.

Due to this construction and the similitude of the triangles, we obtain the conjugate points formula (or equation of the thin lens):

$$\frac{1}{Z} + \frac{1}{Z'} = \frac{1}{D} \tag{15}$$

The image of a point in the scene at distance Z from the lens is produced (in focus) at a distance Z', from the lens, which depends on the depth Z of the point and the focal length D of the lens.

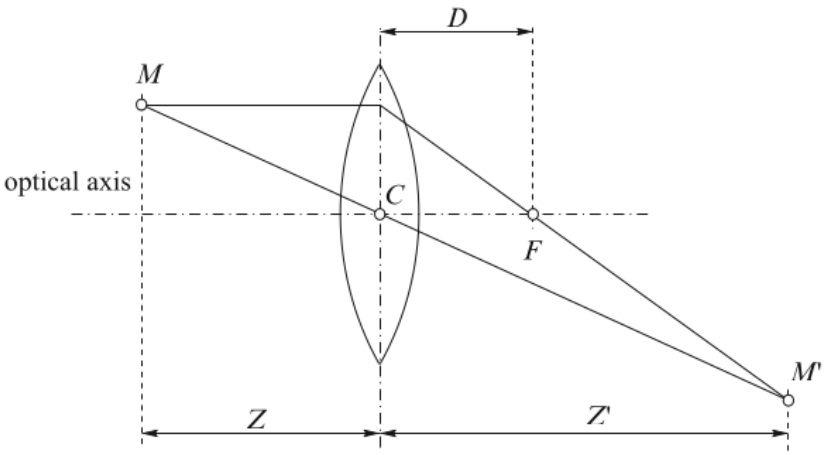


Figure 9: Construction of the conjugate point.

The image of the point M, when in focus, is formed on the image plane at the same point predicted by the pinhole model with the pinhole coincident with the centre of the lens C: in fact, the ray passing through C is common to the two geometric constructions. The other light rays that leave M and are collected by the lens to increase the light reaching M'.

If Equation 15 is not verified one gets a blurred image of the point, that is, a circle which is named **circle of confusion**.

> This is the same effect that would occur by just increasing the size of the pin hole, but is caused by different physical conditions.

The photosensitive elements have a small but finite size.

Remark 1.5. As long as the circle of confusion does not exceed the size of the photosensitive element, the image is in focus. This is due to the **sampling theorem**.

Therefore, there is an interval of depths for which the points are in focus. This range is called **depth of field**, which is inversely proportional to the diameter of the lens; in fact, the pinhole camera has an infinite depth of field. The light that is collected, on the other hand, is directly proportional to the diameter of the lens.

Remark 1.6. The focal length of the lens is a different notion from the focal length of the pinhole camera. Two different notations, f and D, were used on purpose, although, unfortunately, the two quantities have the same name.

1.2 Calibration

We now address the problem of precisely and accurately determining the parameters of the pinhole model (intrinsic and extrinsic), a process that goes by the name of **calibration of the camera**. The idea of the resection methods for calibration is that by knowing the image projections of 3D points of known coordinates (**control points**), it should be possible to obtain the unknown parameters by solving the perspective projection equation. K, R, T can ben obtained by applying the perspective projection equation to the control points and their projections in the image.

1.2.1 Direct linear transform method (Tsai)

Given the correspondences between n non-coplanar control points and their projections in the image, we compute the perspective projection matrix P. The image coordinates of the control points can be obtained manually or by classical image processing methods. From matrix P, we can obtain the intrinsic and extrinsic parameters of the camera by means of a matrix factorization. In order to make the transformation linear, we use homogeneous coordinates for the points in the scene and their projections in the image.

Given a point X in space, $\underbrace{q'}_{x\times 1} = \underbrace{P}_{3\times 4}\ \underbrace{X}_{4\times 1}$ is its projection, where $P = \underbrace{K}_{3\times 3}\ \underbrace{R}_{3\times 3}\underbrace{[I_3 \mid X_0]}_{3\times 4}$.

- $K = \begin{pmatrix} f_x & s & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{pmatrix}$ is the matrix of intrinsic parameters (**5 parameters**), where:
  - f_x and f_y are the focal lengths in the x and y directions, respectively.
  - c_x and c_y are the coordinates of the principal point.
  - s is the skew coefficient.
- R is the rotation matrix (**3 parameters**).
- X_0 is the translation vector (**3 parameters**).

We need to estimate the matrix P, then get K, R, and X_0 from P.

From each control point we obtain two equations since:

$$\begin{pmatrix} u' \\ v' \\ w' \end{pmatrix} = P \begin{pmatrix} U \\ V \\ W \\ T \end{pmatrix} \tag{16}$$

But we can normalize the homogeneous coordinates, we obtain:

$$\begin{pmatrix} u'/w' \\ v'/w' \\ 1 \end{pmatrix} = P \begin{pmatrix} U/T \\ V/T \\ W/T \\ 1 \end{pmatrix} \tag{17}$$

therefore:

$$\begin{pmatrix} x' \\ y' \\ 1 \end{pmatrix} = P \begin{pmatrix} X \\ Y \\ Z \\ 1 \end{pmatrix} \tag{18}$$

we obtain 2 equations for each control point

$$x' = \frac{p_{11}X + p_{12}Y + p_{13}Z + p_{14}}{p_{31}X + p_{32}Y + p_{33}Z + p_{34}} \qquad y' = \frac{p_{21}X + p_{22}Y + p_{23}Z + p_{24}}{p_{31}X + p_{32}Y + p_{33}Z + p_{34}} \tag{19}$$

from which we have to estimate the 11 parameters of P.

> In theory, then, six non-coplanar points are sufficient for the computation of P ; if any four of them are coplanar, the rank of the matrix drops and the DLT algorithm fails.
>
> In practice, it is advisable to use all the available points (the more, the better) to compensate for the inevitable measurement errors.

1.2.1.1 Projection matrix estimation

We know need to estimate the 11 parameters of P, given the 3D coordinates of X_i with $i \in I, |I| \geq 6$ control points, and the observed image coordinates q_i of their projections produced by the uncalibrated camera, $q_i = PX_i, \forall i \in I$.

$$q_i = PX_i = \begin{pmatrix} p_{11} & p_{12} & p_{13} & p_{14} \\ p_{21} & p_{22} & p_{23} & p_{24} \\ p_{31} & p_{32} & p_{33} & p_{34} \end{pmatrix} X_i = \begin{pmatrix} A^T \\ B^T \\ C^T \end{pmatrix} X_i$$

$$q_i = PX_i = \begin{bmatrix} p_{11} & p_{12} & p_{13} & p_{14} \\ p_{21} & p_{22} & p_{23} & p_{24} \\ p_{31} & p_{32} & p_{33} & p_{34} \end{bmatrix} X_i = \begin{bmatrix} A^\top \\ B^\top \\ C^\top \end{bmatrix} X_i \tag{20}$$

in other words

$$\begin{pmatrix} x_i \\ y_i \\ 1 \end{pmatrix} = \begin{pmatrix} u_i \\ v_i \\ w_i \end{pmatrix} = \begin{pmatrix} A^T X_i \\ B^T X_i \\ C^T X_i \end{pmatrix} \tag{21}$$

finally:

$$\begin{aligned} x_i = \frac{u_i}{w_i} = \frac{A^T X_i}{C^T X_i} \quad &\rightarrow \quad x_i C^T X_i - A^T X_i = 0 \\ y_i = \frac{v_i}{w_i} = \frac{B^T X_i}{C^T X_i} \quad &\rightarrow \quad y_i C^T X_i - B^T X_i = 0 \end{aligned} \tag{22}$$

Furthermore, we can rewrite them in order to have the equations with the same variables on the left side:

$$\begin{aligned} x_i X_i^T C - X_i^T A = 0 \quad &\rightarrow \quad -X_i^T A + 0B + x_i X_i^T C = 0 \\ y_i X_i^T C - X_i^T B = 0 \quad &\rightarrow \quad +0A - X_i^T B + y_i X_i^T C = 0 \end{aligned} \tag{23}$$

Up until now we had $P = \begin{pmatrix} A^T \\ B^T \\ C^T \end{pmatrix}$, $P \in \mathbb{R}^{3\times4}$, we rewrite the parameters as a vector $p = \begin{pmatrix} A \\ B \\ C \end{pmatrix} = \text{vec}\ (P^T), p \in \mathbb{R}^{12\times1}$.

The previous equations are now expressible in a more compact form $a_{x_i}^T p = 0$ and $a_{y_i}^T p = 0$,where $a_{x_i}^T = (-X_i^T, 0, x_i X_i^T)$ and $a_{y_i}^T = (0, -X_i^T, y_i X_i^T)$.

Finally, we can collapse the points in a single matrix multiplication:

$$M = \begin{pmatrix} a_{x_1}^T \\ a_{y_1}^T \\ \vdots \\ a_{x_n}^T \\ a_{y_n}^T \end{pmatrix} \rightarrow Mp = 0 \tag{24}$$

In the real world, we will never have $MP = 0$, instead we will have $MP = \omega$, therefore we need to find the vector p that minimizes the error $\|Mp\|_F^2 = \Omega = \omega^T\omega = p^T M^T Mp$. We also want a p such that $\|p\| = 1$, this is to avoid the trivial solution $p = 0$ (which is not useful) and we also fix the scale of the solution by a scalar.

1.2.1.1.1 Least square method

We want to find $\hat{p} = \text{argmin}_p\ \|Mp\|_F^2$ subject to $\|p\| = 1$. This is an algebraic optimization problem, not directly related to geometric entities. We solve it by the **least square method**. Solving a problem of the form $Ax = 0$ with a square matrix A is equivalent to look for the null space of A with $\|x\|^2 = 1, (x \neq 0)$. The eigenvector that corresponds to the smallest eigenvalue of A solves the linear system of equations.

We perform the SVD (Singular Value Decomposition) of M:

$$M = USV^T = \sum_{i=1}^{12} s_i u_i v_i^T \tag{25}$$

where

$$U^T U = I_{12} \quad V^T V = I_{12} \quad S = \text{diag}(s_1, s_2, ..., s_{12}) \text{ with } s_1 \geq s_2 \geq ... \geq s_{12} \tag{26}$$

The last column of V is the eigenvector corresponding to the smallest eigenvalue of $M^T M$, and therefore it is the solution of our problem, $p = v_{12}$. From p we can reconstruct the matrix P and then we can extract the intrinsic and extrinsic parameters of the camera.

1.2.1.1.2 Conditions for a solution

- We assume there are no random or gross measurement errors in the image coordinates of the control points.
- M has rank 11 then, at least 6 control points are needed.

There is no solution if :

- All the control points are sampled on a plane (no 3D information).
- All the control points and the projection center are located on a twisted cubic curve (not very likely).

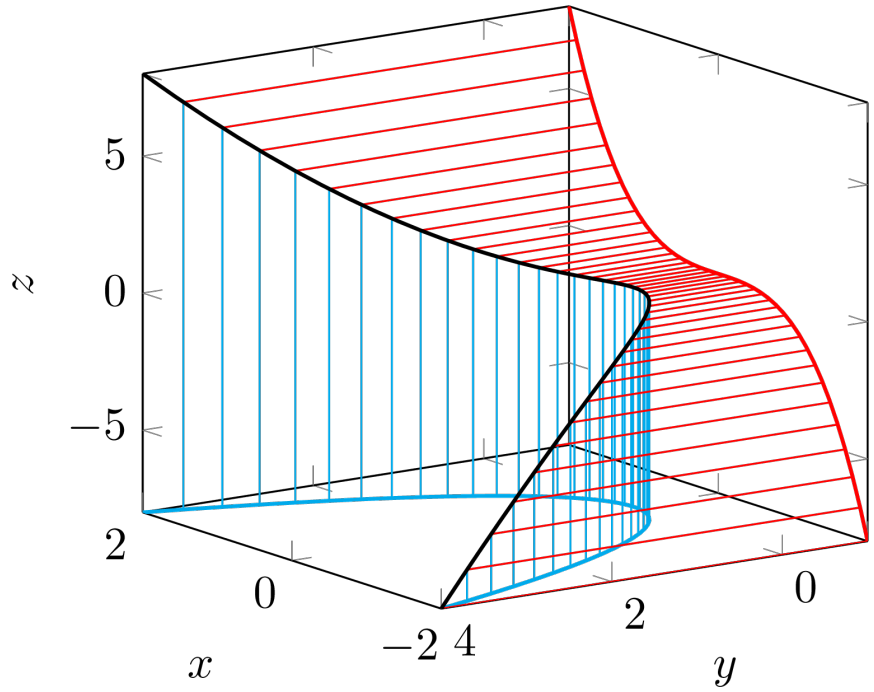


Figure 10: Twisted cubic curve.

1.2.1.2 Parameter extraction

Now that we have P, we can compute X_0, K and R.

$$P = KR[I_3 \mid -X_0] = \left[\underbrace{H_\infty}_{3\times3} \mid \underbrace{h}_{3\times1}\right] \tag{27}$$

We split P by dividing its first 3 columns from the last one.

$$H_\infty = KR \quad h = -KRX_0 \tag{28}$$

Where K is an upper triangular matrix and R is a rotation matrix. R^{-1} also is a rotation matrix, and K^{-1} is also an upper triangular matrix.

Once we have K and R, we can easily compute X_0 by inverting the equation $h = -KRX_0$.

$$X_0 = -H_\infty^{-1} h = R^{-1}K^{-1}KRX_0 \tag{29}$$

But first, we need to find K and R. This can be done by using the **QR decomposition** of H_∞.

Definition 1.7 *(QR decomposition).* The QR decomposition of a square matrix M is a decomposition of the form $M = QR$, where Q is an rotation matrix and R is a triangular matrix.

This decomposition always exists, but it can be not unique.

$$H = QR \rightarrow H_\infty^{-1} = R^{-1}K^{-1} \tag{30}$$

where:

- R^{-1} is the rotation matrix
- K^{-1} is the triangular matrix

these matrixes are obtained by doing the QR decomposition of H_∞^{-1}. We then obtain K and R by inverting K^{-1} and R^{-1}, then we normalize K by dividing it by k_{33} to have $k_{33} = 1$.

1.2.2 Reprojection error

The reprojection error is the Euclidean distance (in pixels) between an observed image point and the corresponding 3D point reprojected onto the camera frame. This give us a quantitative measure of the accuracy of the calibration (ideally 0).

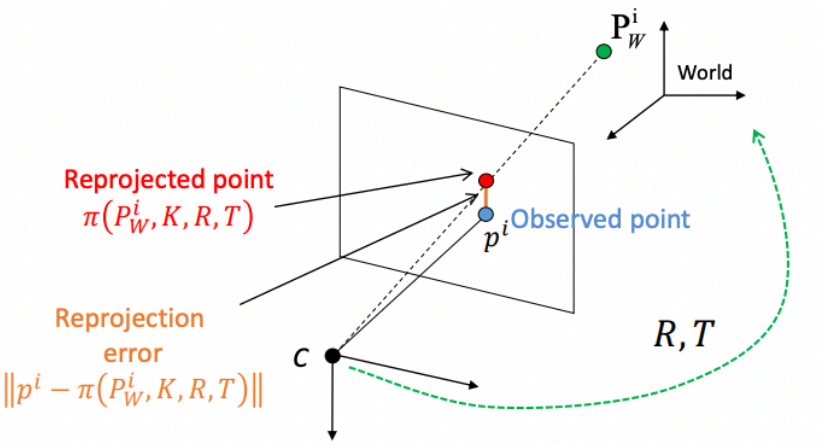


Figure 11: Reprojection of a point.

We can optimize the calibration by minimizing the reprojection error, which is a non-linear optimization problem, in which we also include the lens distortion parameters. This is usually done using an iterative method such as the Levenberg-Marquardt algorithm.

1.2.3 Zhang's method

This method is the state of the art of camera calibration, and is the most used in practice. It is based on the use of **multiple views of a planar calibration pattern**, which is easier to manufacture and use than a 3D calibration object.

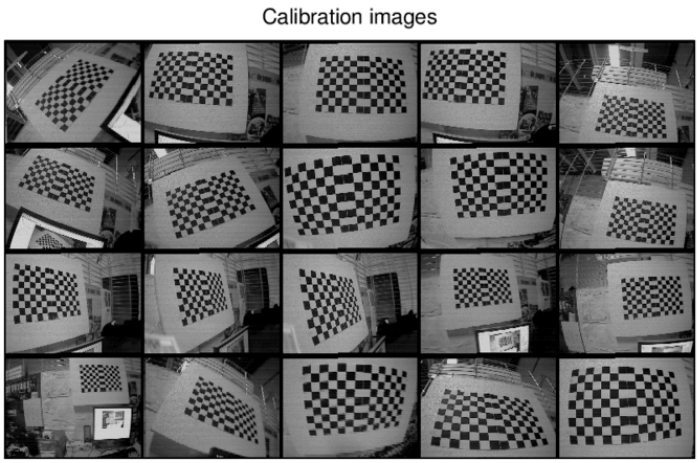


Figure 12: Zhang's method for camera calibration. The calibration pattern is a planar checkerboard, which is easy to manufacture and use.

First, we need to compute where are the corners of the checkerboard (purely image processing problem). This is done because there are many pixels on the corners of the boxed, by interpolating the corners we smooth out the noise.

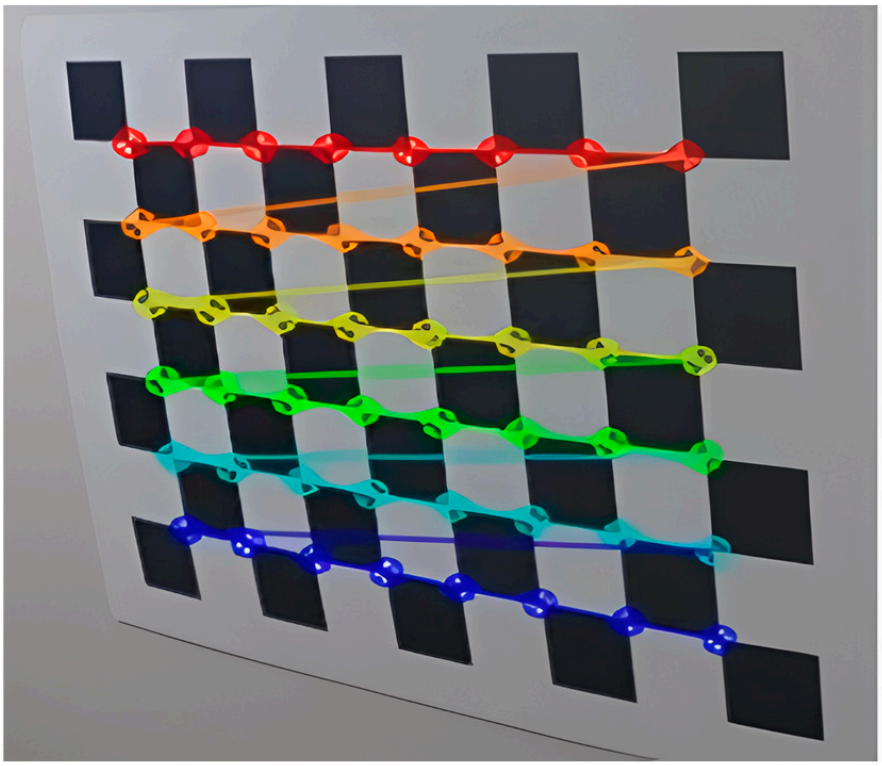


Figure 13: Zhang's method for camera calibration. The corners of the checkerboard are detected in the image.

This method only calculates the intrinsic parameters of the camera.

$$\underbrace{x}_{\substack{3\times1\\ \text{pixel}\\ \text{coordinate}}} = \underbrace{P}_{\substack{3\times4\\ \text{projection}\\ \text{matrix}}} \underbrace{X}_{\substack{4\times1\\ \text{world}\\ \text{coordinate}}} = KR[I_3|\ X_0]X \tag{31}$$

1.2.3.1 Checkerboard trick

Since we are observing a checkerboard, we know the size and structure of the pattern.

We set the world coordinate system to a specific corner of the checkerboard

Remark 1.8. The origin point must lie on the inside of the outer layer of the checkerboard, this is because it avoids the outer white checks where there is no edge.

Since all these points lie on the same plane, in which we take as the origin the corner of a check and the X and Y axis along the edges of such check, then all point have $Z = 0$ as a coordinate. Then, the equation becomes

$$\overbrace{\begin{pmatrix} x \\ y \\ 1 \end{pmatrix}}^{X} = \overbrace{\begin{pmatrix} c & cs & x_H \\ 0 & c(1+m) & y_H \\ 0 & 0 & 1 \end{pmatrix}}^{K} \begin{pmatrix} r_{11} & r_{12} & \cancel{r_{13}} & t_1 \\ r_{21} & r_{22} & \cancel{r_{23}} & t_2 \\ r_{31} & r_{32} & \cancel{r_{33}} & t_3 \end{pmatrix} \overbrace{\begin{pmatrix} X \\ Y \\ \cancel{Z} \\ 1 \end{pmatrix}}^{X} \tag{32}$$

$$\begin{pmatrix} x \\ y \\ 1 \end{pmatrix} = \begin{pmatrix} c & cs & x_H \\ 0 & c(1+m) & y_H \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} r_{11} & r_{12} & t_1 \\ r_{21} & r_{22} & t_2 \\ r_{31} & r_{32} & t_3 \end{pmatrix} \begin{pmatrix} X \\ Y \\ 1 \end{pmatrix}$$

One point generates such equation.

> This is not the general equation of the projection of a point, it works in this **specific** case.

We say that $H = \begin{pmatrix} h_1 & h_2 & h_3 \end{pmatrix} = K\begin{pmatrix} r_1 & r_2 & t \end{pmatrix}$, then for multiple observed points on the checkerboard **from the same image** we have that $\begin{pmatrix} x_i \\ y_i \\ 1 \end{pmatrix} = H \begin{pmatrix} X_i \\ Y_i \\ 1 \end{pmatrix} \quad i = 1, ..., I.$

Then, we can use the same approach to solve the system as in DLT

$$\begin{aligned} a_{x_i}^T h = 0 \\ a_{y_i}^T h = 0 \end{aligned} \tag{33}$$

where

$$\begin{aligned} h = (h_k) = \texttt{vec}\ (H^T) \\ a_{x_i}^T = (-X_i, -Y_i, -1, 0, 0, 0, x_i X_i, x_i Y_i, x_i) \\ a_{x_i}^T = (0, 0, 0, -X_i, -Y_i, -1, y_i X_i, y_i Y_i, y_i) \end{aligned} \tag{34}$$

> We need at least 4 points per plane to estimate H.

1.2.3.2 Extracting the parameters

Once H is known for an image, we have to extract K from it. Unlike DLT, we do not have a rotation matrix so we cannot apply the QR decomposition.

$$H = (h_1 \;\; h_2 \;\; h_3) = \underbrace{\begin{pmatrix} c & cs & x_H \\ 0 & c(1+m) & y_H \\ 0 & 0 & 1 \end{pmatrix}}_{K} \underbrace{\begin{pmatrix} r_{11} & r_{12} & t_1 \\ r_{21} & r_{22} & t_2 \\ r_{31} & r_{32} & t_3 \end{pmatrix}}_{(r_1 \;\; r_2 \;\; t)} \tag{35}$$

$$(r_1 \;\; r_2 \;\; t) = K^{-1}(h_1 \;\; h_2 \;\; h_3)$$
$$\rightarrow r_1 = K^{-1}h_1 \quad r_2 = K^{-1}h_2$$

Since r_1 and r_2 are rows of the rotation matrix, they are orthonormal, i.e.

$$r_1^T r_2 = 0 \text{ and } \|r_1\| = \|r_2\| = 1 \tag{36}$$

Then, we can rewrite Equation 36 by substitution:

$$\begin{aligned} r_1^T r_2 = 0 &\rightarrow \boxed{h_1^T K^{-T} K^{-1} h_2 = 0} \\ \|r_1\| = \|r_2\| = 1 &\rightarrow h_1^T K^{-T} K^1 h_1 = h_2^T K^{-T} K^{-1} h_2 \\ &\rightarrow \boxed{h_1^T K^{-T} K^1 h_1 - h_2^T K^{-T} K^{-1} h_2 = 0} \end{aligned} \tag{37}$$

Define the symmetric and positive definite matrix $B := K^{-T}K^{-1}$ then we rewrite Equation 37:

$$\begin{aligned} h_1^T B h_2 &= 0 \\ h_1^T B h_1 - h_2^T B h_2 &= 0 \end{aligned} \tag{38}$$

If we have B, we can compute K using the **Cholesky decomposition**

$$B = \begin{pmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \\ b_{31} & b_{32} & b_{33} \end{pmatrix} \rightarrow \text{chol}(B) = AA^T \rightarrow A = K^{-T} \tag{39}$$

Define the vector $b = (b_{11}, b_{12}, b_{13}, b_{22}, b_{23}, b_{33})$ of unknowns from the constraints, we construct a system of linear equations $Vb = 0$, where

$$V = \begin{pmatrix} v_{12}^T \\ v_{11}^T - v_{22}^T \end{pmatrix} \text{ with } v_{ij} = \begin{pmatrix} h_{1i}h_{1j} \\ h_{1i}h_{2j} + h_{2i}h_{1j} \\ h_{3i}h_{1j} + h_{1i}h_{3j} \\ h_{2i}h_{2j} \\ h_{3i}h_{2j} + h_{2i}h_{3j} \\ h_{3i}h_{3j} \end{pmatrix} \text{ which are all elements of } H \tag{40}$$

Then for one image, we obtain $\begin{pmatrix} v_{12}^T \\ v_{11}^T - v_{22}^T \end{pmatrix} b = 0$.

Since we need to estimate 6 parameters, at least 3 images are necessary, we stack the vectors in the system. From the system we obtain b, and thus K. This solution can be find in the same way we did for DLT, using the SVD decomposition on B.

1.2.4 Camera localization

This is the problem of determining the pose of a camera (position and orientation) with respect to the world frame, from a set of 3D-2D point correspondences. This is also known as the **Perspective-n-Point (PnP)** problem. We assume the camera is already calibrated, therefore we know the intrinsic parameters. We can use DLT to solve this, but it's suboptimal, we want to find an algebraic solution to the problem. With n=3 points we can solve the problem algebraically, but there are up to 4 solutions, and we need to choose the one that is physically possible (the one in front of the camera). With n=4 points we can solve the problem algebraically, and there is a unique solution.

1.2.5 Robustness to outliers (RANSAC)

In real calibration data, not all detected correspondences are correct. Corner detection can fail, some points can be mismatched, and image noise can perturb localization. These wrong correspondences are called **outliers**.

Classical least-squares estimation (DLT, homography fitting, PnP fitting) is very sensitive to outliers: even a small number of wrong points can significantly bias the estimated parameters.

RANSAC (Random Sample Consensus) is used to make estimation robust:

- Randomly select a minimal subset of correspondences (e.g., the minimum needed to estimate the model).
- Estimate the model parameters from this subset.
- Compute the reprojection error for all correspondences.
- Count how many points are **inliers**, i.e., points with error below a threshold.
- Repeat many times and keep the model with the largest inlier set.
- Re-estimate the final model using all inliers only.

Why we need it in camera calibration:

- It removes the influence of bad checkerboard corner detections or wrong matches.
- It provides a reliable initialization for non-linear refinement (e.g., Levenberg–Marquardt).
- It improves stability of estimated intrinsics/extrinsics, especially with many images and automatic feature extraction.

> RANSAC does not replace optimization: it first identifies a geometrically consistent inlier set, then a final least-squares/non-linear optimization is performed on those inliers.

1.3 Sensing

What differentiates a camera (in the modern sense of the word) from the portable camera obscura of the 17th century is its ability to record the pictures that form on its backplane. We now introduce two different kinds of modern camera sensors.

1.3.1 CCD sensors

Charge-coupled-device (CCD) cameras that were proposed in 1970. A CCD sensor uses a rectangular grid of electron-collection sites laid over a thin silicon wafer to record a measure of the amount of light energy reaching each of them.

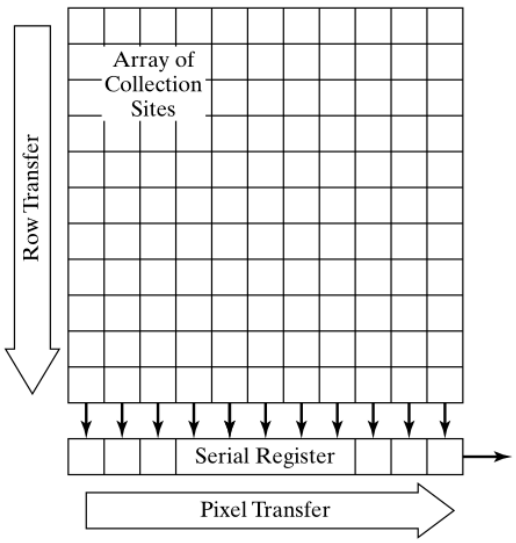


Figure 14: A CCD Device.

Each site is formed by growing a layer of silicon dioxide on the wafer and then depositing a conductive gate structure over the dioxide. When photons strike the silicon, electron-hole pairs are generated (photo-conversion), and the electron are captured by the potential well formed by applying a positive electrical potential to the corresponding gate. The electrons generated at each site are collected over a fixed period of time T.

At this point, the charges stored at the individual sites are moved using charge coupling:

- Charge packets are transferred from site to site by manipulating the gate potentials, preserving the separation of the packets.
- The image is read out of the CCD one row at a time, each row being transferred in parallel to a serial output register with one element in each column.
- Between two row reads, the register transfers its charges one at a time to an output amplifier that generates a signal proportional to the charge it receives.

This process continues until the entire image has been read out. The reading time depends on the use case, varying greatly.

> The digital output of most CCD cameras is transformed internally into an analog video signal before being passed to a frame grabber that constructs the final digital image.

Consumer-grade color CCD cameras essentially use the same chips as black-and-white cameras, except that successive rows or columns of sensors are made sensitive to red, green or blue light often using a filter coating that blocks the complementary light.

The spatial resolution of single-CCD cameras is of course limited, and higher-quality cameras use a beam splitter to ship the image to three different CCDs via color filters.

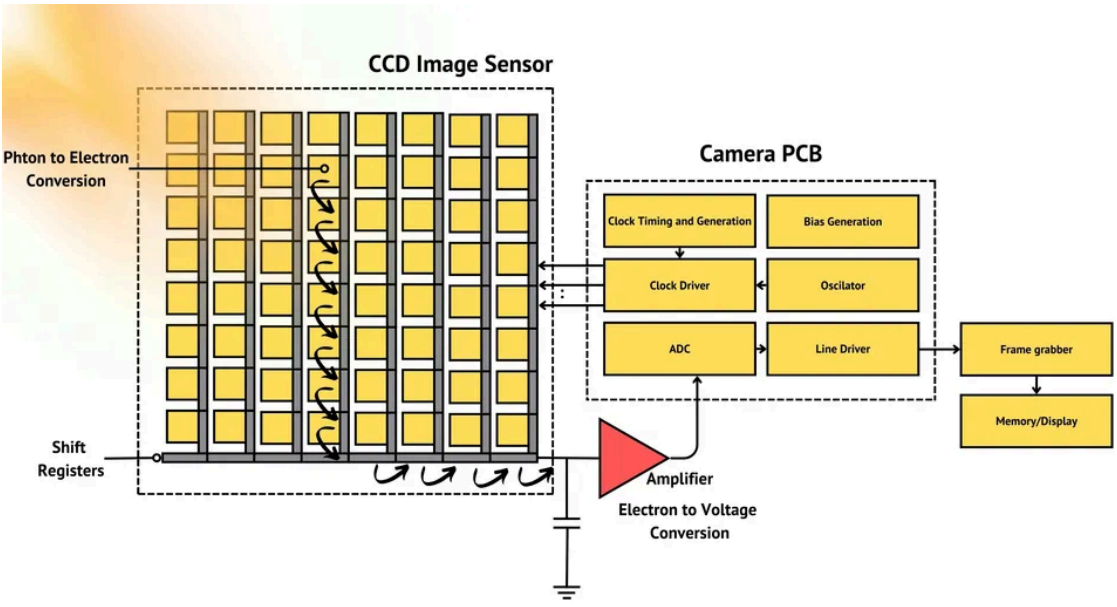


Figure 15: CCD image sensor pipeline

Remark 1.9. The flow of electricity in CCD sensors obviously generates heat, which equals energy. The sensors pick up this heat as noise, therefore the sensors need to be cooled to reduce noise.

1.3.2 CMOS sensors

CMOS sensors are integral to modern digital imaging, particularly in devices like smartphones, digital cameras, and high-speed industrial systems. Unlike CCD sensors, CMOS sensors process image data in a way that allows for faster operation and lower power consumption.

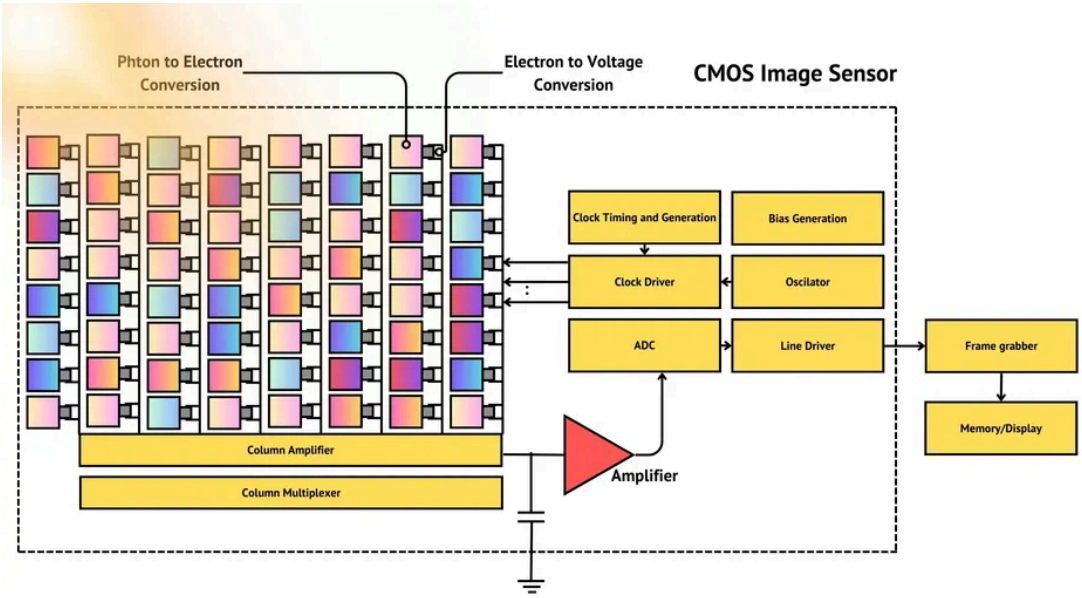


Figure 16: CMOS image sensor pipeline

Just like in a CCD sensor, the process starts when light enters the camera lens and is focused onto the CMOS sensor. The CMOS sensor is made up of pixels arranged in a grid. Each pixel acts as a light detector, capturing photons and converting them into an electrical charge. When photons hit the CMOS sensor, they are absorbed by the photodiodes within each pixel. This absorption generates electron-hole pairs in the silicon substrate, where each photon generates one electron. The number of electrons generated in each pixel is proportional to the intensity of the light striking that pixel. Brighter areas in the image produce more electrons, corresponding to higher brightness levels.

After the charge is converted to voltage at the pixel level, the signal is amplified to make it strong enough for further processing. This amplification occurs within the same pixel, minimizing the distance the signal has to travel, which reduces noise and improves image quality. In many CMOS sensors, the analog voltage signals are converted to digital signals at the column or even the pixel level by an integrated ADC. This localized conversion is one of the reasons CMOS sensors can achieve high-speed readouts.

One characteristic of many CMOS sensors is the **rolling shutter** effect. Because each row or column of pixels is read out sequentially rather than all at once, moving objects can appear distorted. For example, a fast-moving car might look skewed because different parts of the image are captured at slightly different times. However, advanced CMOS sensors can mitigate this effect, and some even employ a **global shutter** to capture all pixels simultaneously, similar to CCD sensors.

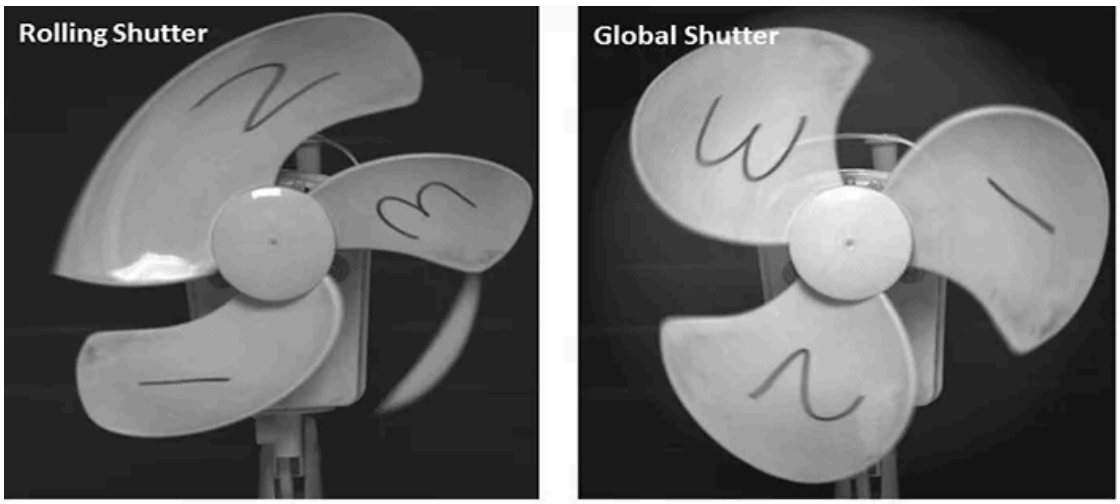


Figure 17: Rolling and global shutter example of the same image.

Optimally we want a global shutter with an exposure time that doesn't lead to motion blur.

1.3.3 Sensor models

For simplicity, we restrict our attention in this section to black-and-white CCD cameras: Color cameras can be treated in a similar fashion by considering each color channel separately and taking the effect of the associated filter response explicitly into account.

The number I of electrons recorded at the cell located at row $r$and column c of a CCD array can be modeled as

$$I(r,c) = T \int_{\lambda} \int_{p \in S(r,c)} E(p,\lambda) R(p) q(\lambda) dp \; d\lambda \tag{41}$$

where T is the electron-collection time and the integral is calculated over the spatial domain $S(r,c)$ of the cell and the range of wavelengths to which the CCD has a nonzero response. In this integral:

- E is is the power per unit area and unit wavelength arriving at the point p
- R is the spatial response of the site
- q is the quantum efficiency of the device

Definition 1.10 *(Quantum efficiency of a device).* The quantum efficiency of a device is the number of electrons generated per unit of incident light energy

In general, both E and q depend on the light wavelength λ, and E and R depend on the point location p within $S(r,c)$.

The output amplifier of the CCD transforms the charge collected at each site into a measurable voltage. In most cameras, this voltage is then transformed into a low-pass-filtered (generally a mean) video signal by the camera electronics with a magnitude proportional to I. The analog image can be once again transformed into a digital one using a frame grabber that spatially samples the video signal and quantizes the brightness value at each image point or pixel (from picture element).

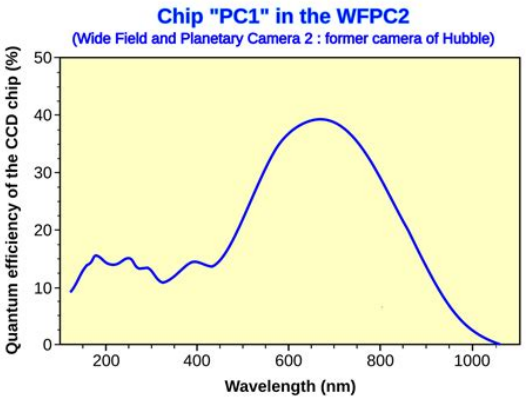


Figure 18: quantum efficiency of Hubble camera 2

In addition to this, we can create a diagram that shows us the relation between the amount of energy incoming into the device (independently of the wavelength) and the "amount" of signal coming out of the device, the so-called **in/out diagram**. Of course, we ideally want it to be linear.

1.4 Digital images

Here we introduce digital images and some operations that can be done on them.

1.4.1 HDR

According to exposure time, length of exposure time and the scene itself some areas of the image can be overexposed, underexposed, or correctly exposed. We want to obtain an image in which all areas are properly exposed. This is called **High Dynamic Range (HDR)** imaging. Of course, the approach changes whether the image is static or dynamic.

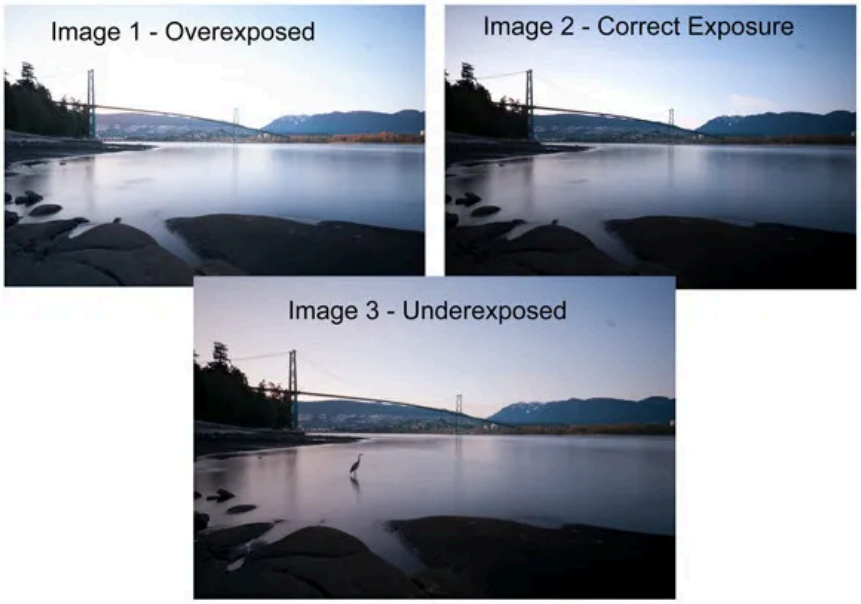


Figure 19: pictures taken with different exposure times of the same scene

There are two main approaches to HDR imaging:

1.4.1.1 Multiple exposure

In this approach, multiple very fast and near exposures are taken on the same scene, with different exposure conditions. These images are combined after being read from the sensor, the algorithm may be heavy and the image can be shown only after the processing is done. This may be acceptable depending on the use case.

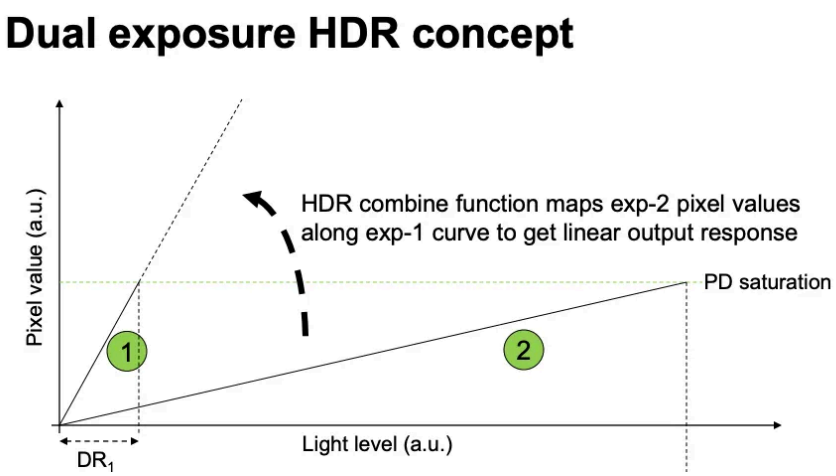


Figure 20: Multiple exposure HDR imaging

1.4.1.2 Single exposure

This works thanks to the fact that in CMOS sensors, a bit of computation can be done on the pixels themselves. Each pixel can be read multiple times during the exposure time (each one has different "filling" times), and the values can be combined to obtain a single value for that pixel. This allows to obtain a single image with a high dynamic range. A problem rises once the image must be read from the sensor, we could have more bits per pixel than the sensor can read. We can linearly map the values into the range of the sensor, but this might be hard to do with a really small amount of transistors per pixel. Typically, the values are mapped into a linear-logarithmic scale, with can be easily implemented in hardware on the pixels.

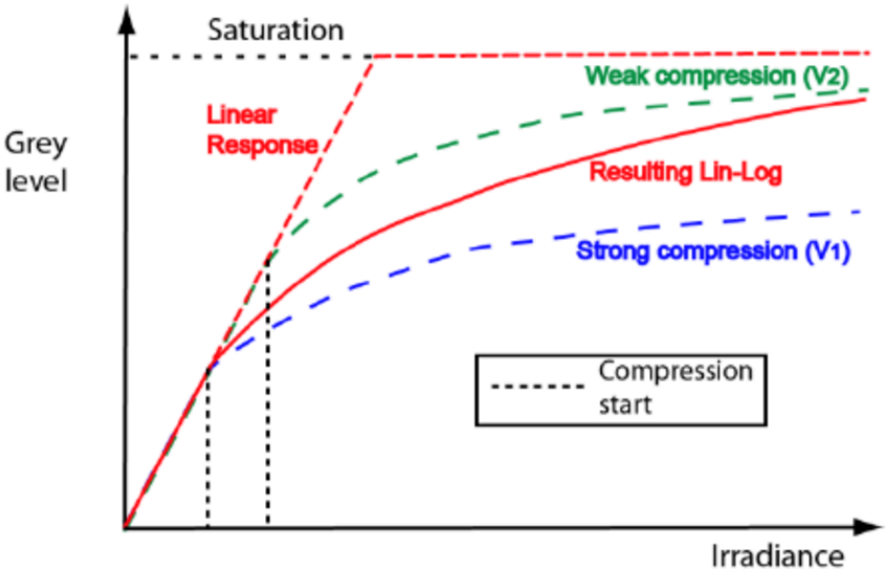


Figure 21: Single exposure HDR imaging

Please note that this introduces a non-linear mapping of the values, which is a change of semantics of the values, and therefore a change of the meaning of the image.

1.4.2 Color cameras

There are different approaches to color cameras:

1.4.2.1 Color filter array

Typically different color channels are captured by different sensors, and the final image is obtained by combining the three channels. This is done by using a color filter array (CFA) that allows each sensor to capture only one color channel (red, green, or blue). The most common CFA is the Bayer filter, which has a pattern of red, green, and blue filters arranged in

a specific way. The raw data from the sensors is then processed using a demosaicing algorithm to reconstruct the full-color image.

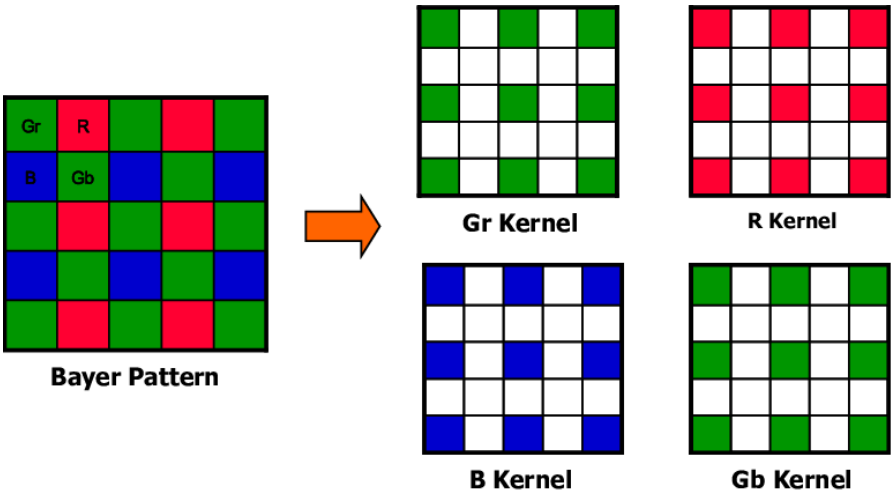


Figure 22: Bayer filter pattern

Quantum efficiency is not defined in the same way for color cameras, since it depends on the wavelength of the light. The colored image is obtained by interpolating the values of the three channels, which are not sampled at the same points. This process is called **demosaicing**. There are many algorithms for demosaicing. The choice of algorithm can affect the quality of the final image, particularly in terms of color accuracy and the presence of artifacts. In general, all demosaicing algorithms introduce artifacts to some level.

Three sensors are sufficient because the color space is three-dimensional, and red, green and blue are a base of the color space.

Remark 1.11. So in practice, when creating a color image, we are sub-sampling the color channels, and then interpolating them to obtain a full-color image.

1.4.2.2 3-CCD cameras

In 3-CCD cameras, the light from the scene is split into three separate beams using a prism or dichroic mirrors, and each beam is directed to a separate CCD sensor that captures one of the primary colors (red, green, or blue). This allows for higher color fidelity and resolution compared to single-sensor cameras with color filter arrays. The three separate images are then combined to create the final color image. 3-CCD cameras are often used in professional video and photography applications. This method avoids the need for demosaicing, as each color channel is captured at full resolution, resulting in better color accuracy and reduced artifacts. But the sensor must be kept in the same position with a micrometer precision, and the prism must be aligned with the sensors. This makes the camera more expensive and less robust to shocks. Also, some of the energy is lost in the prism, and the camera is bigger and heavier than a single-sensor camera.

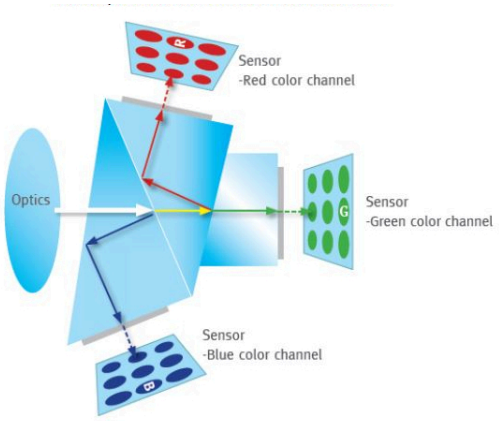


Figure 23: 3-CCD camera

1.4.2.3 Stacked sensors (Foveon)

Three different layers of sensors are stacked on top of each other, with each layer sensitive to a different range of wavelengths (red, green, or blue). This allows for full-color capture at each pixel location without the need for demosaicing. Stacked sensors can provide high color fidelity and resolution, but they can be more complex and expensive to manufacture compared to traditional single-layer sensors with color filter arrays.

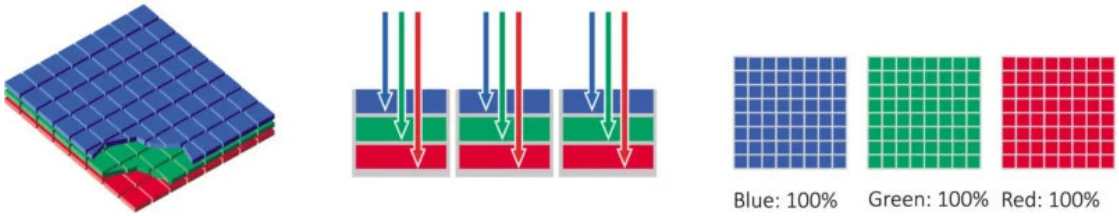


Figure 24: Foveon sensor

1.5 3D cameras

A 3D camera outputs a 4-channel image, with the first three channels being the color channels, and the fourth channel being the depth channel (D). The depth channel contains the distance of each pixel from the camera. This allows to obtain a 3D representation of the scene. Alternatively, in addition to the color channels, the camera can output a 3D point cloud, which is a set of points in 3D space that represent the scene. Each point has a position in 3D space and a color value. This allows to obtain a 3D representation of the scene.

1.5.1 Time-of-flight cameras

We will focus on time-of-flight (ToF) cameras, which are a type of 3D camera that measures the distance to objects in the scene by calculating the time it takes for a signal to travel from the camera to the object and back. ToF cameras **emit a signal** and measure the time it takes for the signal to return after reflecting off objects in the scene. This time measurement is then used to calculate the distance to each point in the scene, creating a depth map.

> Light is fast, too fast...

This time can be measured in two ways: directly as Δt or indirectly after conversion to a phase delay $\Delta\varphi$ through amplitude modulation.

In the first case, also called pulsed wave (PW) LiDAR, the laser source emits a pulse, the receiver records the instant of time when it returns after being reflected by the target and compute the TOF Δt. A simple calculation allows to recover the distance: $r = \frac{c\Delta t}{2}$ where c indicates the speed of light.

The depth resolution of the sensor is related to its ability to measure extremely small timescales. Light travels 1 mm in $3.3ps$, and it is difficult to measure intervals smaller than $5–10ps$.

To obtain a more versatile sensor, a continuous wave (CW) is emitted instead of a pulse; this is an amplitude-modulated radiation with a sine wave, and the phase of the reflected signal is measured.

What we described so far is a laser range finder, a sensor that emits a single beam and therefore obtains distance measurements of only one point in the scene at a time. A laser scanner, instead, uses a beam of laser light to scan the scene in order to derive the distance to all visible points.

1.6 Defects in images

Here we introduce some defects that can be present in images. Defects can be caused by the sensor, the lens, or the environment.

1.6.1 Defects caused by the sensor

1.6.1.1 Blooming

Blooming occurs when the light source illuminating a collection site is so bright that the charge stored at that site overflows into adjacent ones.



Figure 25: CCD blooming and smear. (a) A CCD captured image with blooming and smear, (b) another scene image with blooming and smear.

1.6.1.2 Smear

Smear denotes the undesired bright sections above and below the bright light source, caused by charges' accumulation of the light source during the vertical transfer process.

1.6.2 Defects caused by the lens

The field of view of a camera is the portion of scene space that actually projects onto the retina of the camera. It is not defined by the focal length alone, but also depends on the effective area of the retina (e.g., the area of film that can be exposed in a photographic camera, or the area of the CCD sensor in a digital camera.

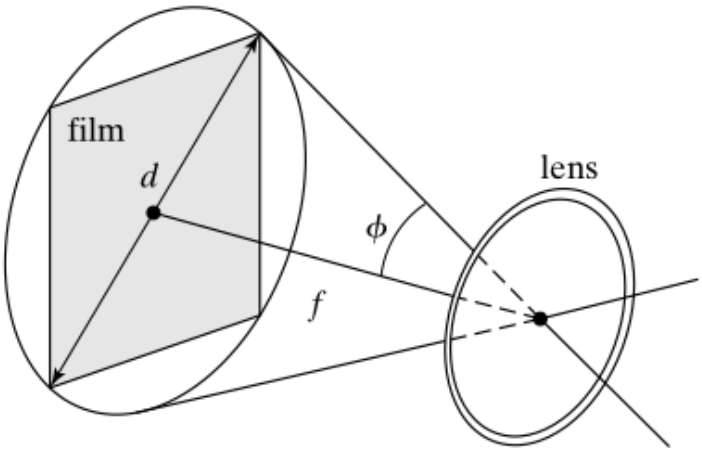


Figure 26: Field of view of a camera

Lenses can introduce distortions in the image, which generally increase with the distance from the optical axis. Therefore, it's better to put the sensor well inside the circle in which the lens projects the image. The most common distortions are:

- **Barrel distortion**: the image magnification decreases with distance from the optical axis, causing straight lines to appear curved outward. This is common in wide-angle lenses.
- **Pincushion distortion**: the image magnification increases with distance from the optical axis, causing straight lines to appear curved inward. This is common in telephoto lenses.

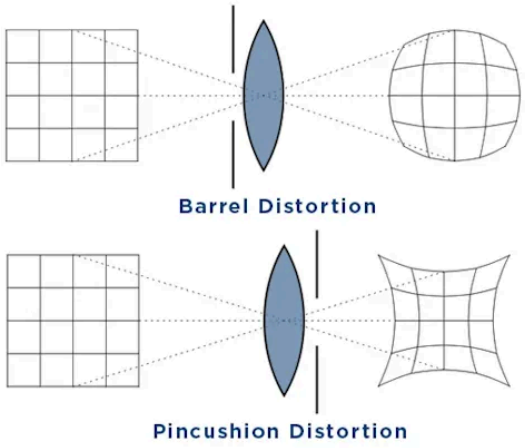


Figure 27: Lens distortion

We can also discriminate between radial and tangential distortion. Radial distortion is caused by the shape of the lens, while tangential distortion is caused by the misalignment of the lens elements.



Figure 28: Radial and tangential distortion

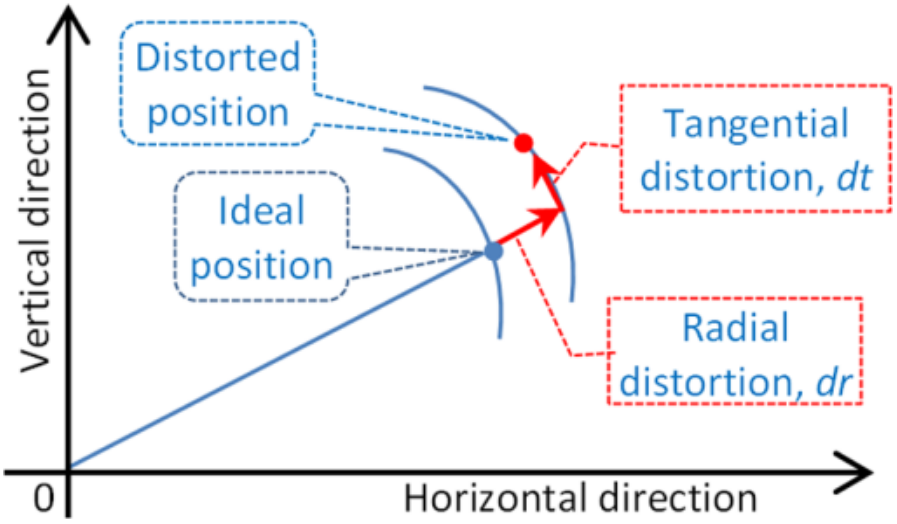


Figure 29: Definitions of radial and tangential distortion on the 2-D plane

Note that both radial and tangential are a **function of the radius**, that is the distance from the optical axis (principal point). Therefore, they are both zero at the optical axis and increase with distance from it.

1.6.2.1 Chromatic aberration

Chromatic aberration is a type of distortion in which there is a failure of a lens to focus all colors to the same convergence point. It occurs because lenses have different refractive indices for different wavelengths of light, causing them to bend light of different colors by different amounts. This results in a misalignment of colors in the image, leading to color fringing around the edges of objects, especially in high-contrast areas. Chromatic aberration can be minimized by using achromatic lenses, which are designed to bring two wavelengths (typically red and blue) into focus in the same plane, or by using apochromatic lenses, which bring three wavelengths into focus in the same plane.

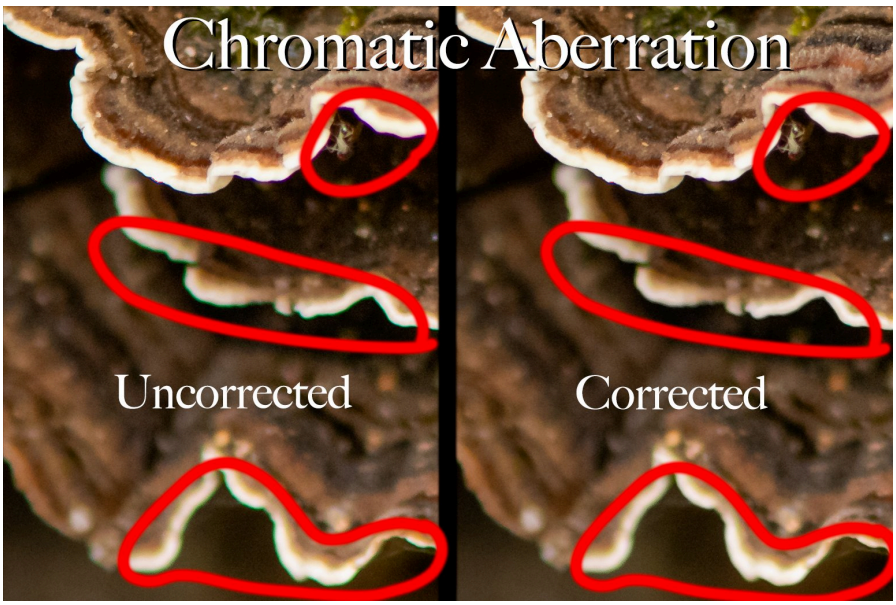


Figure 30: Chromatic aberration example

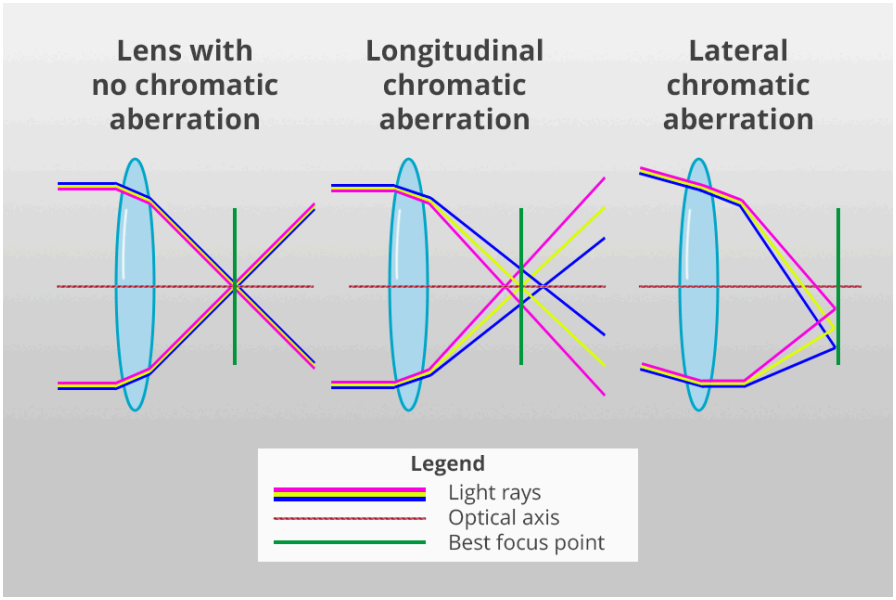


Figure 31: Chromatic aberration types

1.6.2.2 Vignetting

Light beams emanating from object points located off-axis are partially blocked by the various apertures (including the individual lens components) positioned inside the lens to limit aberrations. This phenomenon, called **vignetting**, causes the brightness to drop in the image periphery. Vignetting may pose problems to automated image analysis programs, but it is not as important in photography thanks to the human eye's remarkable insensitivity to smooth brightness gradients.

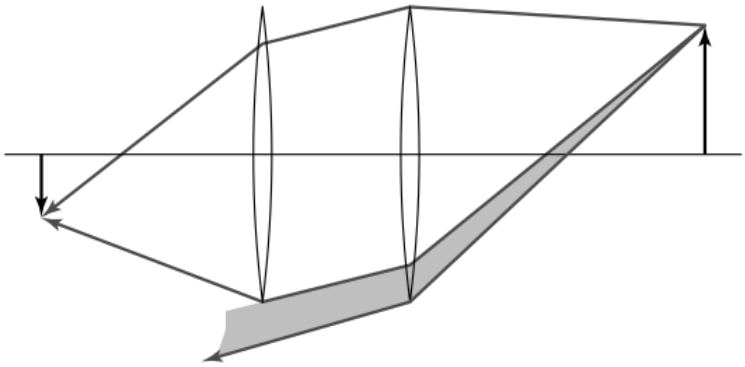


Figure 32: Vignetting effect in a two-lens system. The shaded part of the beam never reaches the second lens. Additional apertures and stops in a lens further contribute to vignetting.



Figure 33: Example of vignetting in a photograph. The corners of the image are darker than the center.

2 Stereoscopy

We want to perceive the world in three dimensions, in order to do this, we need more than one image of the same scene. This is because there is one degree of freedom that is lost when we project a 3D scene onto a 2D image (unless having absolute information about the image). This degree of freedom is the depth, which is the distance of an object from the camera. In order to recover this information, we need to have at least two images of the same scene, taken from different viewpoints. This is the basic idea behind stereoscopy, which is the technique of using two images to create a perception of depth.

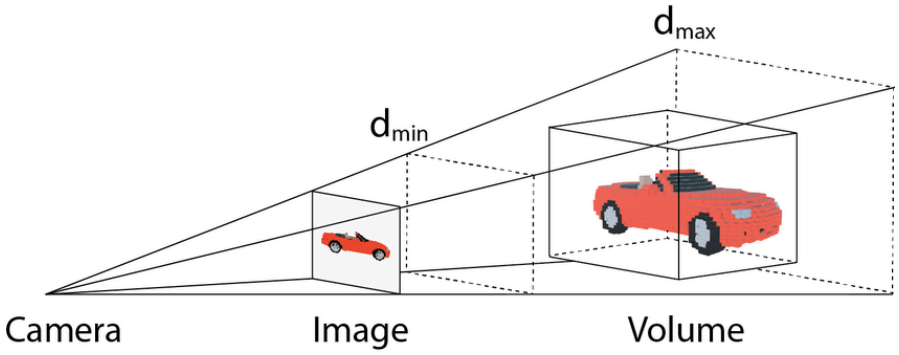


Figure 34: During image formation with a single camera, the depth information is lost.

In human terms, this problem might be solved if additional information is provided (previous experience etc.), but in computer vision, we need to find a way to recover this information from the images themselves. This is where stereoscopy comes into play, as it allows us to recover depth information from images taken from different viewpoints.

Two images alone are not enough, some processing is needed, for example the human brain uses the two images captured by different points of view (the two eyes) simultaneously and process them (in the chiasm).

Actually, there is no real requirement for using only two images, different techniques can be used to recover depth information from multiple images, but the basic idea is the same: using multiple images to recover depth information. Alternately, we can use a **single camera** and **move it to different positions** to capture multiple images of the same scene. This is known as structure from motion (SfM) and is a common technique in computer vision for 3D reconstruction. Here we focus on the case of two images, which is the most common and simplest case, and we will see how to recover depth information from them.

Consider two images, which we call primary and secondary, taken from different viewpoints. From these, we want to obtain a depth image (or point cloud), which is a 3D reconstruction of the scene.

> In a depth image, each pixel contains information about the **distance of the corresponding point in the scene from the camera**. In a point cloud, each point represents a 3D point in space, and the collection of points represents the 3D structure of the scene.

2.1 Geometric model of stereoscopy

In general, we have

- two projection planes Π' and Π''
- their corresponding projection centers C' and C'' and image center O' and O''
- a fixation point O (if any)
- a point P in the scene, which is projected onto the two planes as P' and P''.

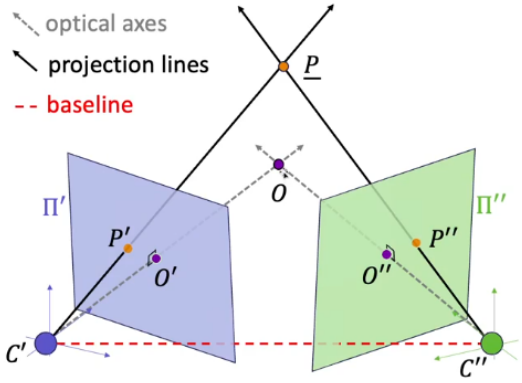


Figure 35: Stereoscopic geometry.

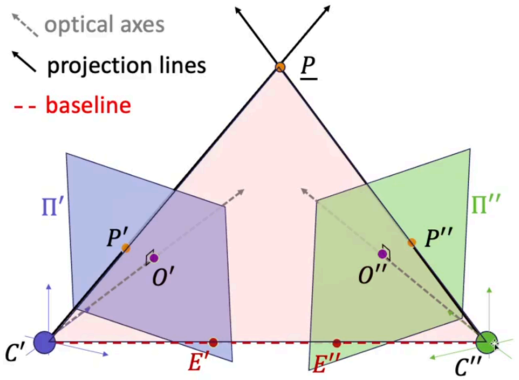


Figure 36: epipolar plane.

The plane defined by the two projection centers C' and C'' and the point $\underline{P}$ is called the **epipolar plane** (P, C', C''). The points E' and E'' are called the **epipoles**, and they are the projections of the fixation point onto the two projection planes. The line connecting the two epipoles is called the **baseline**. This plane, can also be defined without $\underline{P}$, by using the two projections (P', C', C'') or (P'', C', C''). This is useful when we know the coordinates of the projections, but not the coordinates of the point in the scene.

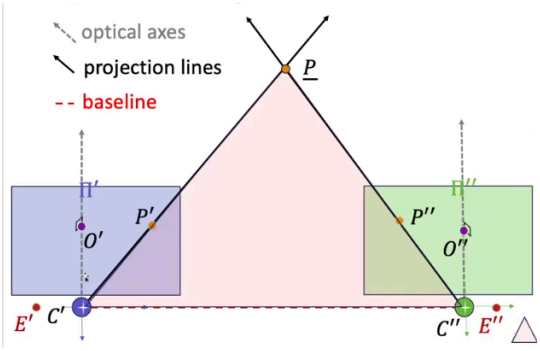


Figure 37: Stereoscopic geometry in standard form.

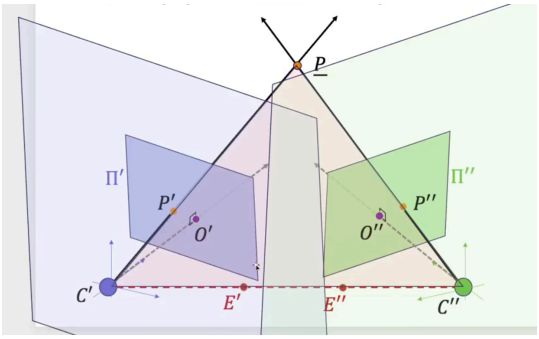


Figure 38: Note that the epipoles are not necessarily inside the projection planes, this is not particularly a problem.

The situation shown in figure is called the **standard form** of stereoscopy (or extreme geometry). In this case, the two optical axes are parallel, and the two projection planes are coplanar. We can also convert any stereoscopic geometry into the standard form by applying a suitable rotation and translation to the two projection planes. This is called **rectification** and it is a common preprocessing step in stereoscopic image processing. In order to do this, we need to know the intrinsic and extrinsic parameters of the two cameras, which can be obtained through camera calibration.

There are two main problems to face in stereoscopy:

- **Correspondence (stereo matching)**: given a point in one image P', find the corresponding point P'' in the other image which is the projection of the same point in the scene $\underline{P}$.
- **Reconstruction**: given the two projections P' and P'', find the 3D coordinates of the point in the scene $\underline{P}$.

Of these two problems, the correspondence problem is the most challenging, as it requires finding the correct match between the two images. Regarding reconstruction, if the cameras are calibrated and we know the intersection between $\overline{C'P'}$ and $\overline{C''P''}$, we can easily compute the 3D coordinates of the point in the scene $\underline{P}$. in the case of errors, noise etc. the two lines may not intersect (they are skew), in this case we can find the point that minimizes the distance from both lines, which is a common approach in computer vision.

2.2 Overview of the stereoscopic pipeline

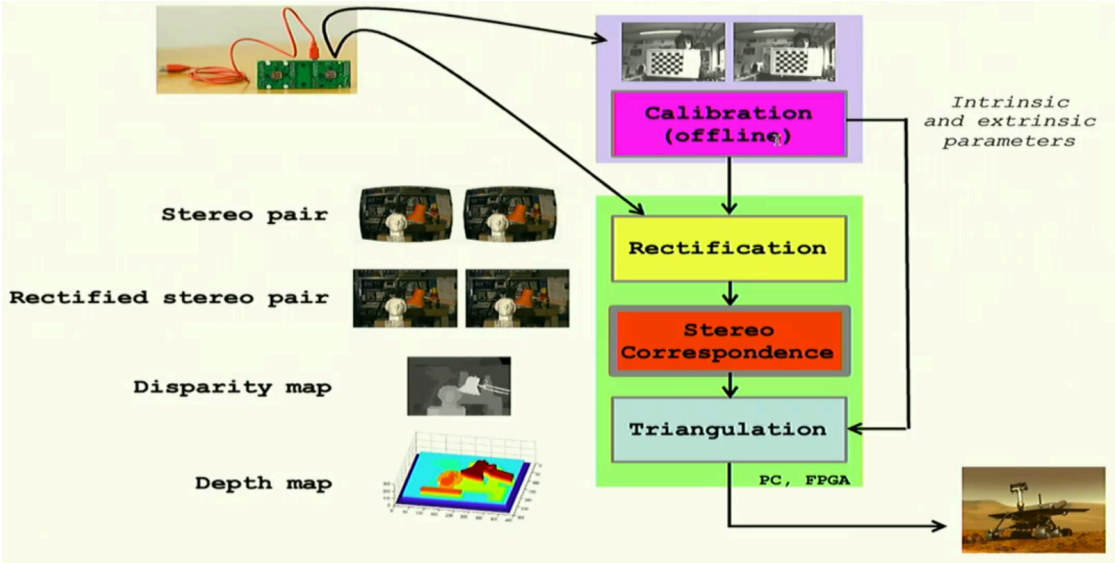


Figure 39: General pipeline of a stereoscopic system.

Remark 2.1. In stereo, calculating the extrinsic parameters means calculating the intersection of the projection lines. Also keep in mind that the world reference frame must be the same for the two cameras. e.g. in Zheng calibration the checkboard must be seen by both cameras, and the world reference frame is the checkboard reference frame.

2.2.1 Rectification

Using the information from the calibration step:

1. remove the distortion from the lenses
2. turn the stereo pair in standard form (rectification)

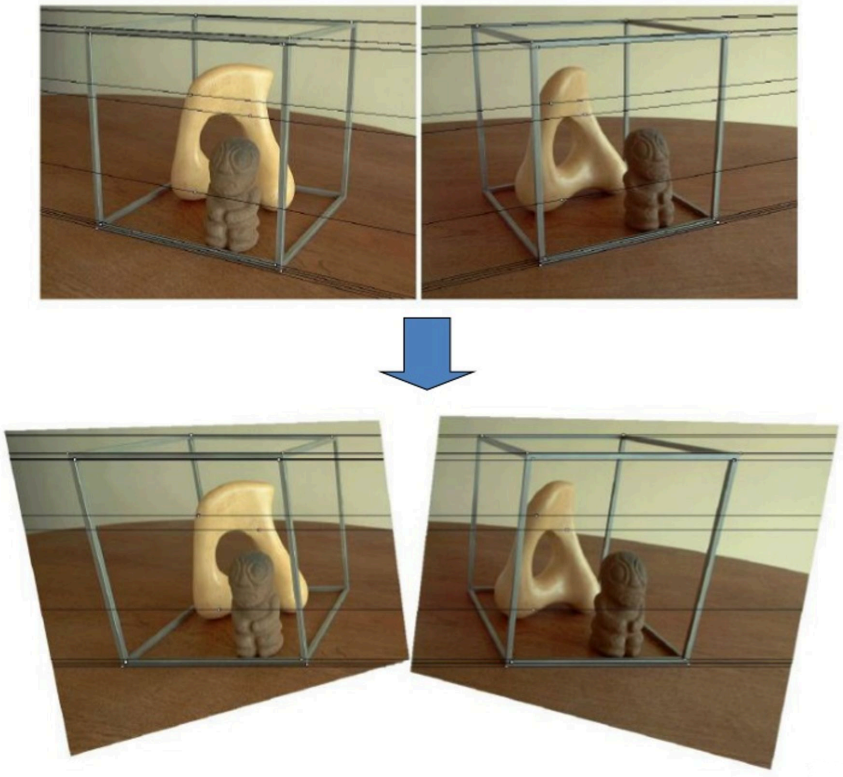


Figure 40: Rectification of a stereo pair.

2.2.2 Correspondence

1. find the corresponding points in the two images (stereo matching)

2. calculate the disparity for each point and output the disparity map

The disparity map may be more or less dense, which also depends on the depth of the scene.

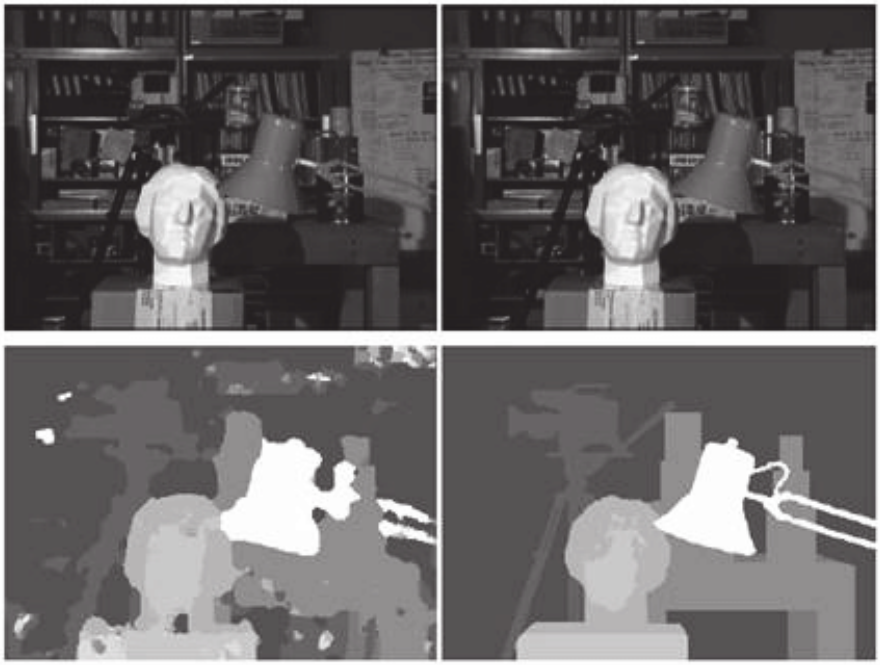


Figure 41: Disparity map of a scene.

2.2.2.1 Difficulties of stereo matching

These factors make the correspondence problem difficult to solve:

- Photometric distorsions and noise
- Specular surfaces
- Foreshortening: might be difficult to uniquely identify a point if nearby points are projected in the same position
- Perspective distortions
- Uniform/ambiguous regions
- Transparent surfaces
- Occlusions: some points are visible in one image but not in the other

2.2.3 Triangulation

Given the disparity map, the baseline and the focal length (which comes from calibration), triangulation computes the position of the correspondence in the 3D space, and outputs a point cloud or a depth map.

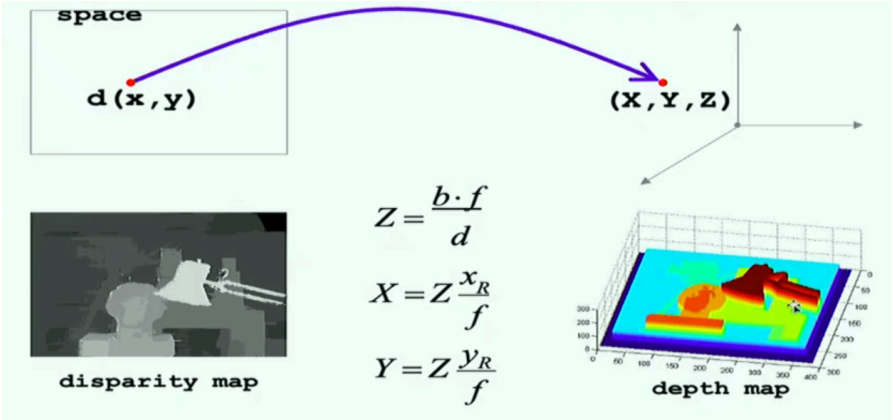


Figure 42: Triangulation of a point in the scene.

2.3 The correspondence problem (stereo matching)

In order to associate two points in the two planes, we need to find a way to match them. As points (in a geometrical sense), they are not distinguishable. For this reason we use a neighborhood of pixels around the point. This is based on the fundamental hypothesis of stereoscopy, that is that corresponding points in the two images have similar neighborhoods. This is called the **similarity assumption**. This hypothesis is not always true, as there are many situations where it fails, for example in the case of occlusions, reflections, or textureless regions. In these cases, we need to use additional information to find the correct match, such as color, texture, or motion.

This is large problem because of the large number of possible matches. For each image we extract a set of salient points $S' = \{P'_1, P'_2, ..., P'_n\}$ and $S'' = \{P''_1, P''_2, ..., P''_m\}$, or indexes in the list of pixels. The combinatorial problem is of size $n \times m$ and is hard to solve efficiently. After the matching we will have pairs of corresponding points $\left(P'_i, P''_j\right)$, of which we will call the left one the primary point and the right one the secondary point. The primary point is the one that we will use to find the corresponding point in the other image, which is the secondary point.

2.3.1 The role of the baseline

Consider the length of $\overline{C'C''}$, $\overline{C'\underline{P}}$ and $\overline{C''\underline{P}}$ the smaller $\overline{C'C''}$ the more similar are the two images, therefore the correspondence problem is easier to solve. This is because the two images are taken from similar viewpoints, and therefore the projections of the same point in the scene are more likely to be similar. This is called the **baseline** of the stereoscopic system, and it is an important parameter that affects the difficulty of the correspondence problem. On the other hand, if the baseline is too small, the depth information is lost, and the quality of the reconstruction is reduced.

In general, the larger $\overline{C'C''}/\overline{C'\underline{P}}$ (or $\overline{C'C''}/\overline{C''\underline{P}}$), the more precise is the reconstruction.

Since the shape generated by lines is a triangle, we call the process of finding the depth information from the two images **triangulation**. The larger the baseline, the more accurate is the triangulation, but the more difficult is the correspondence problem. Therefore, there is a trade-off between the baseline and the difficulty of the correspondence problem.

2.3.2 The reconstruction error

The **reconstruction error** does not depend only on the distance form the baseline, but is also a function of the baseline. The larger the baseline, the less elongated, with respect to the depth, is the triangle $C'C''P$, and therefore the more accurate the reconstruction.

> Given a baseline, the depth of the error grows **quadratically** with the depth

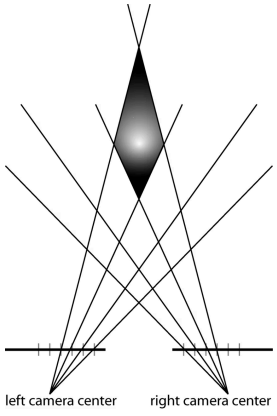


Figure 43: The reconstruction error grows quadratically with the depth.

Also, the probability distribution function of the error can be approximated by a normal (usually the first order of Taylor), but the accuracy of a normal approximation decreases with the depth. This is a known problem in computer vision and many solutions have been proposed, some of which we will see later. Sometimes, a skewed normal may be a better approximation.

2.3.3 Disparity

The **disparity** is the difference in the position of the projections of the same point in the two images.

Definition 2.2 *(Epipolar line).* The **epipolar line** is the intersection of the epipolar plane with the projection plane. It is the line along which the corresponding point must lie in order to satisfy the epipolar constraint.

Definition 2.3 *(Epipolar constraint).* The **epipolar constraint** states that the corresponding point in the other image must lie on the epipolar line (this is because we can construct the epipolar plane from the two camera positions and the principal point on the camera plane). This is a fundamental constraint in stereoscopy, as it reduces the search space for the correspondence problem from 2D to 1D.

In the general case (i.e. not in rectified cameras):

- camera axes are not parallel
- epipolar lines are not horizontal with the same v (or y) coordinate

Disparity is a vector function with two components $\Delta_u(u,v), \Delta_v(u,v)$, each is a function of the coordinates (u,v) of the primary point, the corresponding point of (u_a, v_a) will have coordinates $\left(u_a + \Delta_{u(u,v)}, v_a + \Delta_{v(u,v)}\right)$.

In rectified cameras, disparity has one component $\Delta_u(u,v)$, and the corresponding point of (u_a, v_a) will have coordinates $\left(u_a + \Delta_{u(u,v)}, v_a\right)$.

Of course, it is biunivocally related to the depth. In general, disparity is higher for points that are closer to the camera, and lower for points that are farther away. This is because the projections of closer points will be more separated in the two images, while the projections of farther points will be closer together.

Given a stereo rig with baseline b and focal length f, the range field of the system is constrained by the disparity range $[d_{\min}, d_{\max}]$.

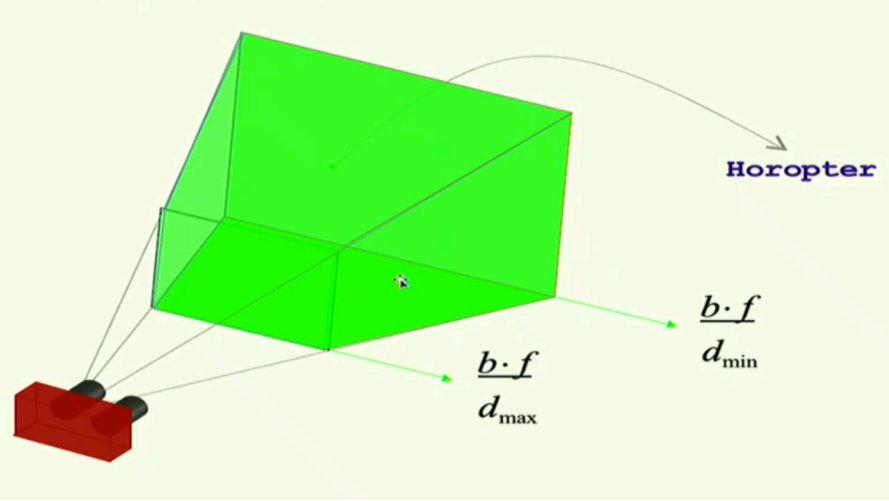


Figure 44: The disparity range constrains the depth range of the system.

But how can we represent the disparity of a pixel? Just by the nature of digitalization, the value will be discrete, then, we can define the plane of all the points with such disparity (all planes will be parallel). This poses a limitation in the ability to reconstruct a scene, as we can only reconstruct points that lie on these planes.

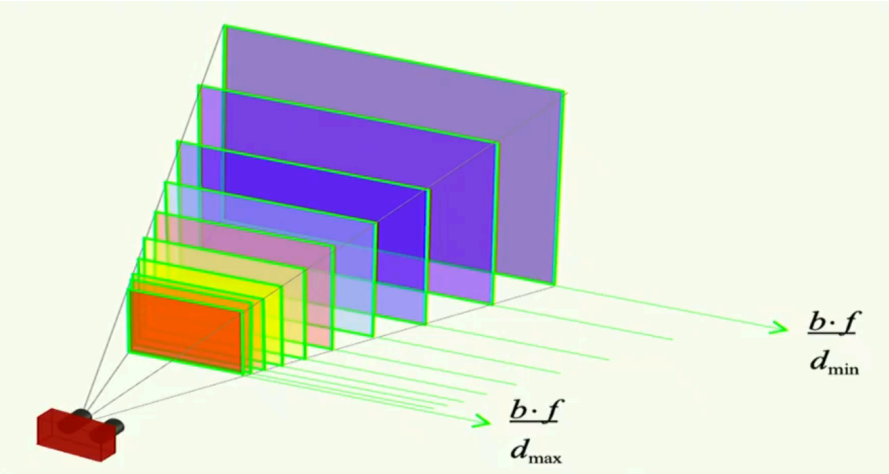


Figure 45: The disparity planes are parallel and discrete, which limits the ability to reconstruct a scene.

As you can see in Figure 45, the distance between the planes is not constant, but it is a function of the depth. This means that the closer the point is to the camera, the more accurate the reconstruction will be, while the farther away the point is, the less accurate the reconstruction will be.

2.3.4 Algorithms for stereo matching

Most stereo algorithms perform (a subset of) the following steps:

1. Matching cost computation: for each pixel in the primary image, compute a cost function that measures how well it matches with pixels in the secondary image along the corresponding epipolar line.
2. Cost aggregation: aggregate the matching costs over a local neighborhood to reduce noise and improve robustness.
3. Disparity computation: select the disparity value that minimizes the aggregated cost for each pixel
4. Disparity refinement: apply post-processing techniques to refine the disparity map, such as median filtering, left-right consistency check, and sub-pixel interpolation.

We can classify the algorithms based on the type of cost function used, the size of the local neighborhood, and the type of optimization used to compute the disparity map. Some common types of algorithms are:

- Local methods: do steps 1-3, simple winner-takes-all (WTA) approach, fast but less accurate, sensitive to noise and occlusions.
- Global methods: $1(\rightarrow 2) \rightarrow 3$ with global (or semi-global) reasoning.

We can also discriminate between pixel-based and feature-based methods, depending on whether they operate on individual pixels or on extracted features (e.g., corners, edges, etc.). Feature-based methods are generally more robust to noise and occlusions, but they may not provide dense disparity maps.

2.3.4.1 Naive

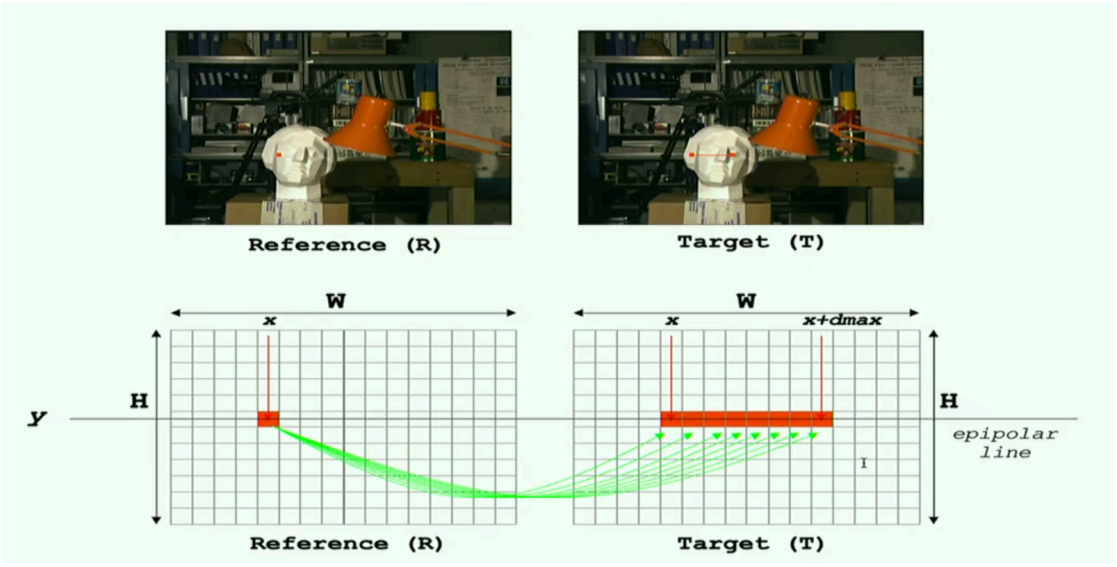


Figure 46: Naive local approach for stereo matching.

In the naive local approach, for each point of interest in the primary image, we search for the corresponding point in the secondary image along the epipolar line. We compute a matching cost (e.g., sum of absolute differences, normalized cross-correlation) for each candidate point and select the one with the lowest cost as the match. Of course we exploit the epipolar constraint to reduce the search space from 2D to 1D. Regarding the matching cost, it uses the Sum of Absolute Differences (SAD): computes the sum of absolute differences between the pixel intensities of the two pixels. Really simple but it is REALLY sensitive to noise (*useless*).

2.3.4.2 Window-based

If we want to keep a **local approach**, in order to increase the signal to noise ration (SNR), the matching costs are aggregated over a support windows.

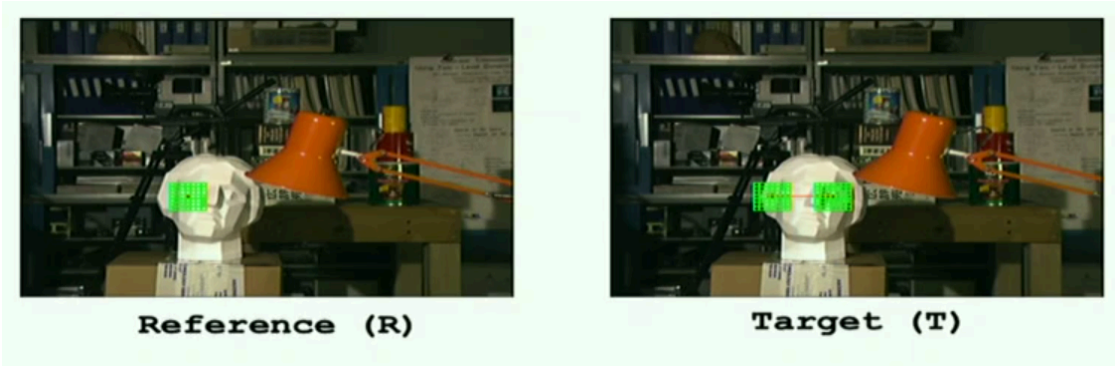


Figure 47: Support window for cost aggregation.

But how do we choose the size of the support window? If it is too small, it will be sensitive to noise, if it is too large, it will be sensitive to depth discontinuities. Therefore, there is a trade-off between noise sensitivity and depth discontinuity sensitivity. We can also use adaptive support windows, which change size and shape based on the local image content, to improve the matching accuracy.

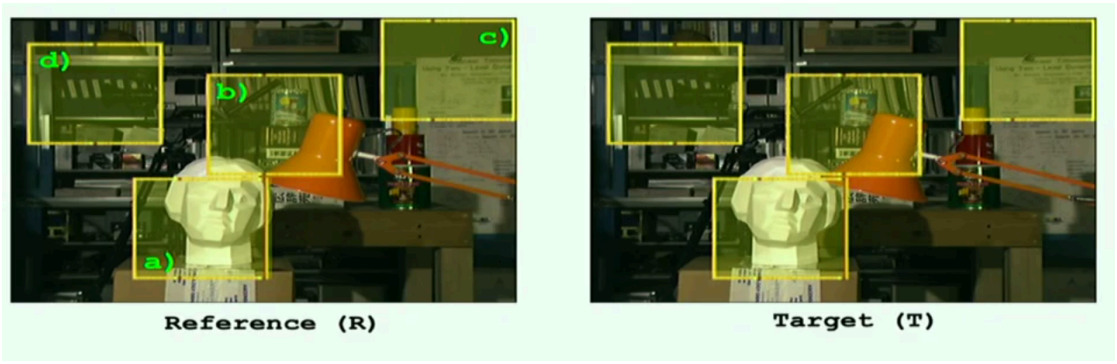
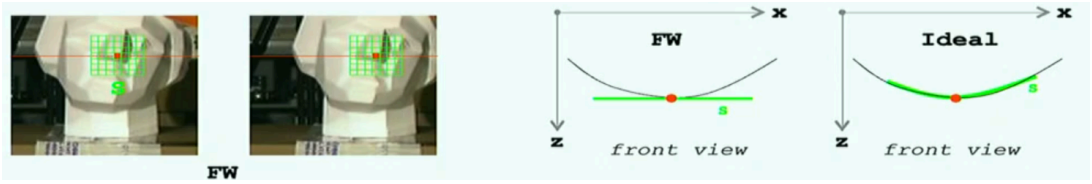


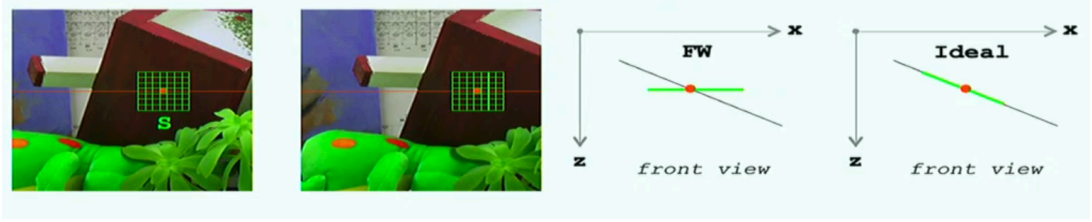
Figure 48: Fixed support window for cost aggregation.

Fixed windows (with WTA reasoning) fails in most points :

- implicitly assumes frontal parallel surfaces



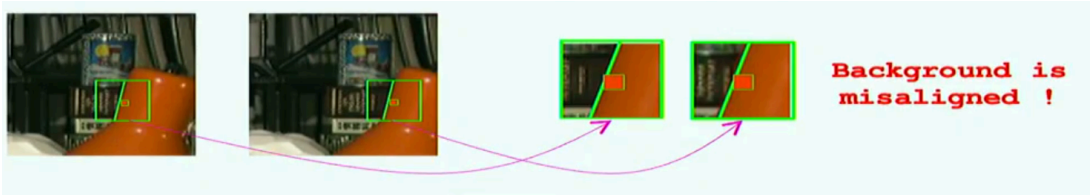
Often violated in pratice



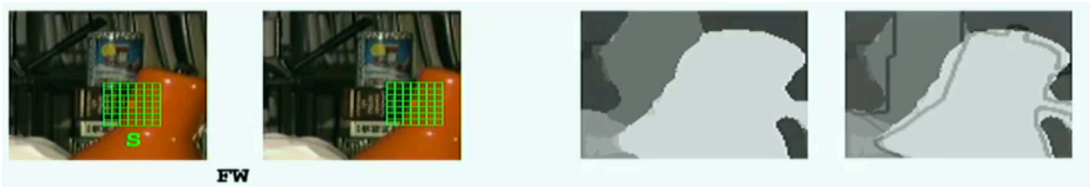
Almost all state of the art cost aggregation strategies rely on the assumption that all the points belonging to the support share the same disparity.

- ignores depth discontinuities

  Implicitly assuming fontral parallel surfaces in the real scene is violated near depth discontinuities.



Aggregating the matching costs of the two populations at different depths (aligned foreground and misaligned background (outliers)) results in the typical inaccurate localization of depth borders



SOA cost aggregation strategies aim at shaping the support in order to include only points with the same (unknown) disparity. In practice, the size of the support is done empirically.

- does not deal explicitly with uniform areas

- does not deal explicitly with repetitive patterns

2.3.4.3 Global

If we want to use a global (or semi-global) approach, we can search for the disparity assignment that minimizes a certain cost function over the stereo pair.

$$E(d) = E_{\text{data}}(d) + E_{\text{smooth}}(d) \tag{42}$$

2.4 Epipolar geometry

We now move on to study the geometry of stereo in its full generality. This will enable us to clarify what information is needed in order to perform the search for corresponding elements only along image lines. First of all, we need to establish some basic notations.

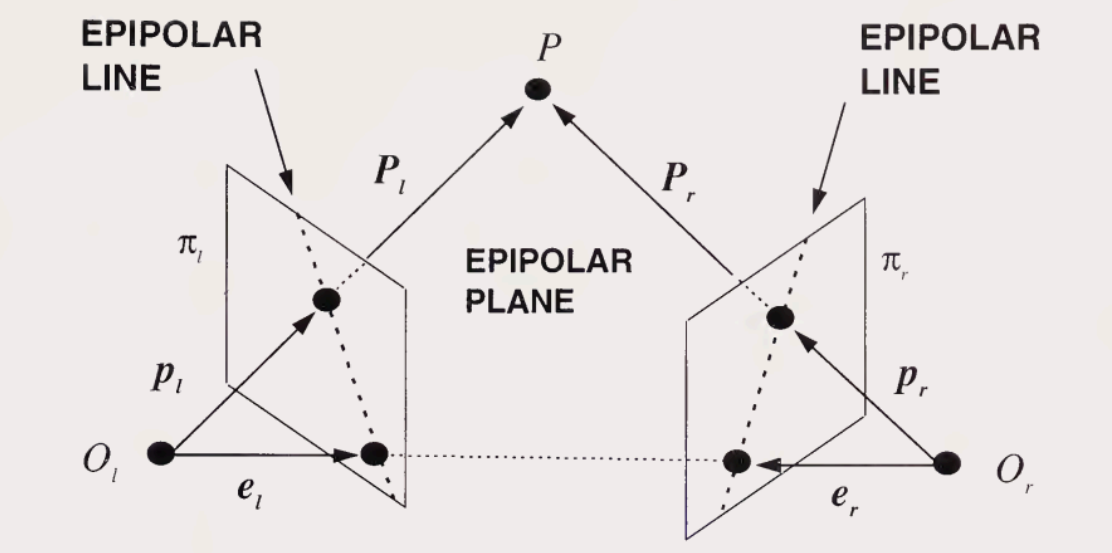


Figure 53: Epipolar geometry.

Figure 53 shows:

- two pinhole cameras
- their projection centers O_l and O_r
- and image planes π_l and π_r
- focal lengths are denoted by f_l and f_r.

As usual, each camera identifies a 3-D reference frame, the origin of which coincides with the projection center, and the Z-axis with the optical axis. The vectors$P_l = [X_l, Y_l, Z_l]^T$ and $P_r = [X_r, Y_r, Z_r]^T$ refer to the same 3-D point, P, thought of as a vector in the left and right camera reference frames respectively. The vectors $p_l = [x_l, y_l, z_l]^T$ and $p_r = [x_r, y_r, z_r]^T$ refer to the projections of P onto the left and right image plane respectively, and are expressed in the corresponding reference frame. Clearly, for all the image points we have $z_l = f_l$ or $z_r = f_r$.

The reference frames of the left and right cameras are related via the extrinsic parameters. These define a rigid transformation in 3-D space, defined by a translation vector, $T = (O_r, O_l)$, and a rotation matrix R. Given a point P in space, the relation between P_l and P_r is therefore

$$P_r = R(P_l - T) \tag{43}$$

We denote the left and right epipole by e_l and e_r, respectively. By construction, the left epipole is the image of the projection center of the right camera and vice versa.

Remark 2.4. If the line through the centers of projection is parallel to one of the image planes (i.e. we are in a rectified configuration), the corresponding epipole is the point at infinity of that line

The relation between a point in 3-D space and its projections is described by the usual equations of perspective projection.

The practical importance of epipolar geometry stems from the fact that the plane identified by P, O_i, and O_r, called epipolar plane, intersects each image in a line, called epipolar line.

Consider the triplet T, p_l, and p_r. Given p_l, P can lie anywhere on the ray from O_l through p_l. But, since the image of this ray in the right image is the epipolar line through the corresponding point, p_r, the correct match must lie on the epipolar line (**epipolar constraint**).

Remark 2.5. Since all rays include the projection center by construction, this also proves that all the epipolar lines go through the epipole.

So, if we determine the mapping between points on, say, the left image and corresponding epipolar lines on the right image, we can restrict the search for the match of p_l along the corresponding epipolar line. The search for correspondences is thus reduced to a 1-D problem. Alternatively, the same knowledge can be used to verify whether or not a candidate match lies on the corresponding epipolar line.

Remark 2.6. This is usually a most effective procedure to reject false matches due to occlusions.

2.4.1 The essential matrix

The equation ol the epipolar plane through P can be written as the coplanarity condition of the vectors P_l, T, and $P_l - T$, or

$$(P_l - T)^T T \times P_l = 0 \tag{44}$$

By the relation defined in Equation 43, we can rewrite this as

$$\left(R^T P_r\right)^T T \times P_l = 0 \tag{45}$$

A vector product can be written as a multiplication by a rank-deficient matrix, we can write

$$T \times P_l = S P_l$$
$$\text{where } S = \begin{pmatrix} 0 & -T_z & T_y \\ T_z & 0 & -T_x \\ -T_y & T_x & 0 \end{pmatrix} \tag{46}$$

Then, we can write the coplanarity condition as

$$P_r^T E P_l = 0 \text{ with } E = RS \tag{47}$$

> By construction, S has always rank 2.

The matrix E is called the **essential matrix** and establishes a natural link between the epipolar constraint and the extrinsic parameters of the stereo system.

By dividing by $Z_r Z_l$ we can write the epipolar constraint in terms of the image coordinates p_r and p_l, as

$$p_r^T E p_l = 0 \tag{48}$$

Please note that if one point is known, we have a system with 2 coordinates parameters from the other point (the third one is f) and the unknowns from the essential matrix (5 parameters). Therefore, from the system we obtain a line, which is the epipolar line. Ep_l is the projective line in the right plane, u_r, that goes through p_r and the epipole e_r.

$$u_r = E p_l \tag{49}$$

Notice that the whole discussion used coordinates in the camera reference frame, but what we actually measure from images are pixel coordinates. Therefore, in order to be able to

make profitable use of the essential matrix, we need to know the transformation from camera coordinates to pixel coordinates, that is, the intrinsic parameters. This limitation is removed in the next section, but at a price.

2.4.2 The fundamental matrix

We now show that the mapping between points and epipolar lines can be obtained from corresponding points only, with no prior information on the stereo system. Let M_l and M_r be the matrices of the intrinsic parameters of the left and right cameras, respectively. if $\overline{p}_l$ and $\overline{p}_r$ are the pixel coordinates of p_l and p_r in camera coordinates, we have

$$\begin{aligned} p_l &= M_l^{-1}\overline{p}_l \\ p_r &= M_r^{-1}\overline{p}_r \end{aligned} \qquad 50$$

By substituting these relations into Equation 48, we can write the epipolar constraint in terms of pixel coordinates as

$$\overline{p}_r^T F \overline{p}_l = 0 \text{ where } F = M_r^{-T} E M_l^{-1} \qquad 51$$

We call F the **fundamental matrix** of the stereo system. As with Ep_l, $F\overline{p}_l$ can be thought of as the equation of the projective epipolar line, $\overline{u}_r$, that correspond to the point $\overline{p}_l$, or

$$\overline{u}_r = F\overline{p}_l \qquad 52$$

The difference between u_r and $\overline{u}_r$ and between the essential and fundamental matrices, is that the fundamental matrix is defined in terms of pixel coordinates, the essential matrix in terms of camera coordinates. Consequently, if you can estimate the fundamental matrix from a number of point matches in pixel coordinates, you can reconstruct the epipolar geometry with no information at all on the intrinsic or extrinsic parameters.

2.4.3 Computing the essential and fundamental matrices: the 8-point algorithm

Of the various methods possible, the eight-point algorithm is by far the simplest. We consider here the fundamental matrix only. The idea behind the eight-point algorithm is very simple. Assume that you have been able to establish n point correspondences between the images. Each correspondence gives you a homogeneous linear equation as in Equation 51 for the nine entries of F, we obtain an homogeneous linear system. If you have at least eight correspondences ($n > 8$) and the $n$points do not form degenerate configurations, the nine entries of F can be determined as the nontrivial solution of the system. Since the system is homogeneous, the solution is unique up to a signed scaling factor. If one uses more than eight points, so that the system is overdetermined, the solution can once again be obtained by means of SVD related techniques (like in camera calibration). If A is the system's matrix and $A = UDV^T$, the solution is the column of V corresponding to the only null singular value of A.

The estimated fundamental matrix is almost certainly nonsingular (which is a requirement for a valid fundamental matrix). We can enforce the singularity constraint by adjusting the entries of the estimated matrix F.

A Mathematical concepts

Here we present some related mathematical concepts that are required for the course.

A.1 Roto-translation

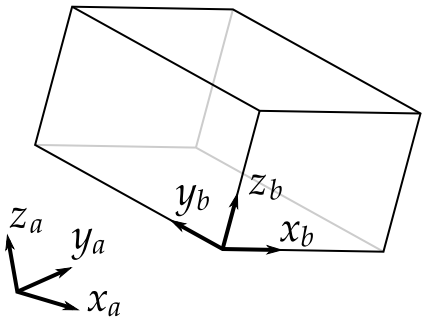


Figure 54: Rigid body attached frame. The position and orientation of a rigid body can be described by attaching a frame to it.

Consider a rigid body as of figure Figure 54, rigid means that the points of the body do not change their relative position. The position and orientation of a rigid body can be described by attaching a frame to it. The coordinates of the points of the object are expressed in the attached frame (x_b, y_b, z_b) and we would want to express them in the reference frame (x_a, y_a, z_a).

In many scientific fields, the **degrees of freedom** of a system is the number of parameters of the system that may vary independently. For example, a point in the 3D space has three degrees of freedom for translation: its coordinates; In 3D space, n points have $3n$ degrees of freedom for translation. If the object is rigid, that is not the case, because the points of the object cannot move independently. In fact, a rigid body has 6 degrees of freedom: 3 for translation and 3 for rotation.

The position and orientation of a rigid body can be described by a roto-translation matrix T that transforms the coordinates of the points of the object expressed in the attached frame (x_b, y_b, z_b) to the reference frame (x_a, y_a, z_a): The union of the position and the orientation of a rigid body is called the **pose** of the rigid body.

A.2 Euler angles

Euler angles describe the orientation of a rigid body in space through three successive intrinsic rotations:

- **Precession (φ):** Rotation around the initial z-axis by an angle φ. The system (x, y, z) becomes (x', y', z'), with $z' = z$.
- **Nutation (θ):** Rotation around the intermediate x'-axis by an angle θ. The system becomes (x'', y'', z''), with $x'' = x'$.
- **Proper rotation (ψ):** Rotation around the intermediate z''-axis by an angle ψ. The final system is (x''', y''', z'''), with $z''' = z''$.

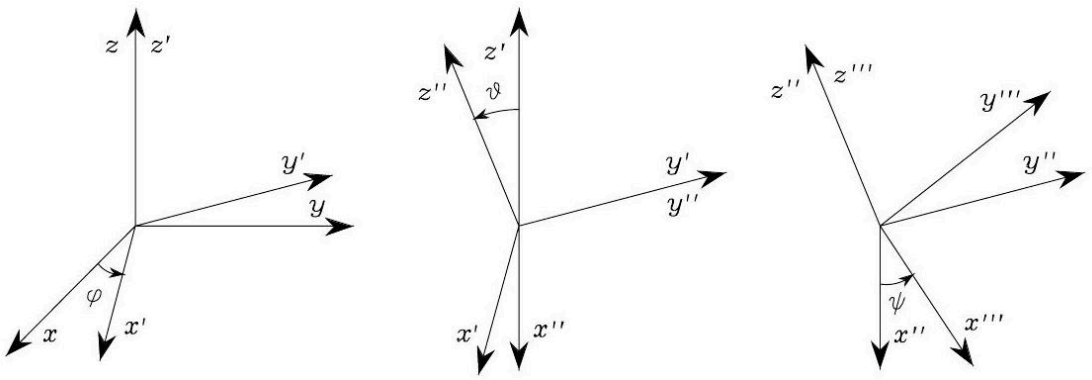


Figure 55: Euler angles. The orientation of a rigid body in space can be described through three successive intrinsic rotations: precession, nutation and proper rotation.

A.3 Roll, pitch and yaw angles

Roll, pitch and yaw angles describe the orientation of a rigid body in space through three successive extrinsic rotations: Roll, pitch and yaw.

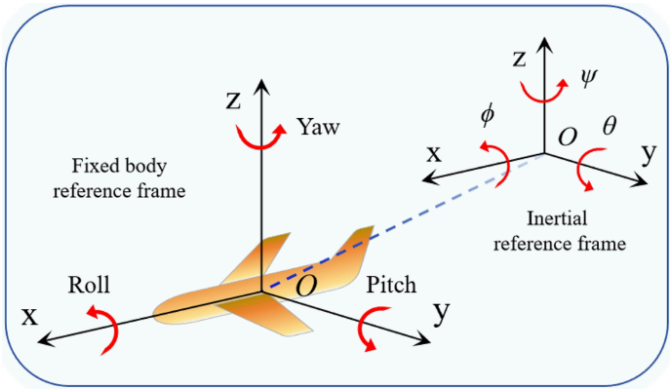


Figure 56: Roll, pitch and yaw angles.

A.4 Roto-translation matrix

Let's combine the translation and rotation of a rigid body in a single transformation matrix T called roto-translation matrix, which represents the relative pose between the relative frame of the object and the reference frame.

Take for example a frame O-zyx, we rotate it around the z-axis by an angle α, we can say that α is the value of one of the degrees of the change of orientation between the two frames.

$$x' = \begin{pmatrix} \cos\alpha \\ \sin\alpha \\ 0 \end{pmatrix} \quad y' = \begin{pmatrix} -\sin\alpha \\ \cos\alpha \\ 0 \end{pmatrix} \quad z' = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \tag{53}$$

Then, the rotation matrices are defined as:

$$\begin{aligned} R_z(\alpha) &= \begin{pmatrix} \cos\alpha & -\sin\alpha & 0 \\ \sin\alpha & \cos\alpha & 0 \\ 0 & 0 & 1 \end{pmatrix} \\ R_y(\beta) &= \begin{pmatrix} \cos\beta & 0 & \sin\beta \\ 0 & 1 & 0 \\ -\sin\beta & 0 & \cos\beta \end{pmatrix} \\ R_x(\gamma) &= \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos\gamma & -\sin\gamma \\ 0 & \sin\gamma & \cos\gamma \end{pmatrix} \end{aligned} \tag{54}$$

We postmultiply the coordinates of a point expressed in the frame O-zyx by the rotation matrix to obtain the coordinates of the point expressed in the rotated frame O-$z'y'x'$. If we compose the rotations, we can compute one after the other, for example, we can first rotate around the z-axis and then around the y-axis:

$$\begin{pmatrix} x'' \\ y'' \\ z'' \end{pmatrix} = R_y(\beta)R_z(\alpha)\begin{pmatrix} x \\ y \\ z \end{pmatrix} \tag{55}$$

We can represent translation and rotation in a single transformation matrix T called roto-translation matrix, which represents the relative pose between the relative frame of the object and the reference frame. To do this, we can use homogeneous coordinates, which allow us to represent both rotation and translation in a single matrix. So given a point $p = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$ we have $\tilde{p} = \begin{pmatrix} w_1 \\ w_2 \\ w_3 \\ w_4 \end{pmatrix}$ with

$$w_1 = \frac{x}{w_4}, w_2 = \frac{y}{w_4}, w_3 = \frac{z}{w_4} \tag{56}$$

and we can express the roto-translation as:

$$\tilde{p}^0 = A_1^0\tilde{p}^1 \tag{57}$$

with

$$A_1^0 = \begin{pmatrix} R_1^0 & o_1^0 \\ 0^T & 1 \end{pmatrix} \tag{58}$$

where R_1^0 is the rotation matrix from frame 1 to frame 0, and o_1^0 is the translation vector of the origin of frame 1 with respect to frame 0. The rotation gives three degrees of freedom, while the translation gives three more degrees of freedom, for a total of six degrees of freedom for a rigid body.